

美赛经验谈

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01

基本介绍

Brief Introduction

建立食物网模型

- 包含生产者、消费者及物质循环（土壤）。
- 考虑农业周期、季节性变化、化学药剂使用对各层级生物量及系统稳定性的影响。

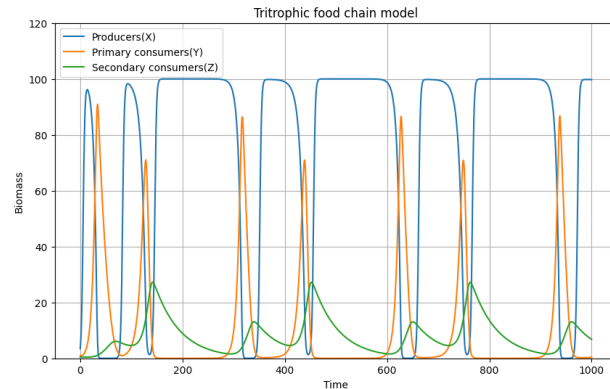
— : 模型一
— : 模型二
— : 模型三

模拟物种回归

- 引入两个新物种，重新考虑问题一

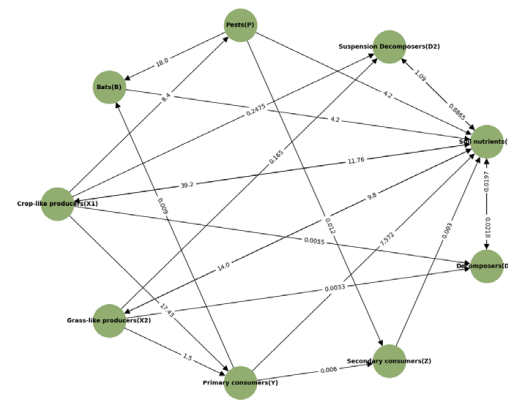
分析人类决策的影响

- 若农民减少或停止使用除草剂，分析对生态系统稳定性的影响
- 如何通过引入有益物种（蝙蝠【指定】、蜜蜂【自选】）恢复生态平衡？
- 若转向有机农业，系统分析对生态系统、作物健康、经济效益等方面的影响如何？



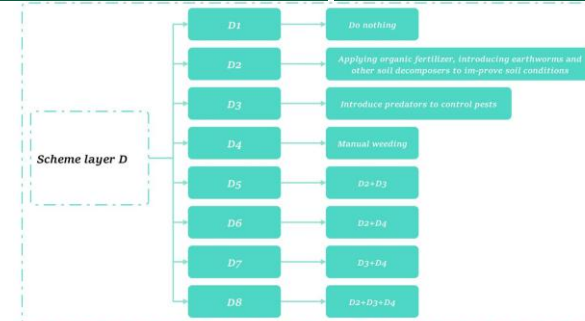
模型一

多层次建模，求解各层级生物量的动态变化



模型二

稳定性评估，分析不同措施对生态网络稳定性的影响



模型三

决策性评价，找到实现绿色农业的最佳方案



“模型一” 主要工作：由易到难，逐步建立三层次微分方程模型，求解各层级生物量的动态变化。既考虑自然农业周期，又考虑人工农药效果。

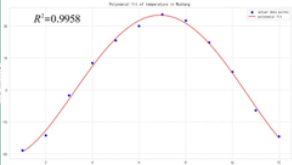
① 数据来源真实。数据引用自《Employing dynamic model to study food web stability and possible regime shifts in Somme Bay》

② 模型假设合理，论证充分。



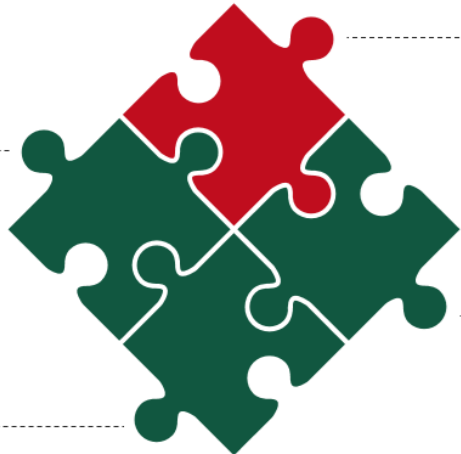
农业周期

将一年划分为生长期和休耕期



气候因子

利用黑龙江五常市真实气温数据拟合温度曲线

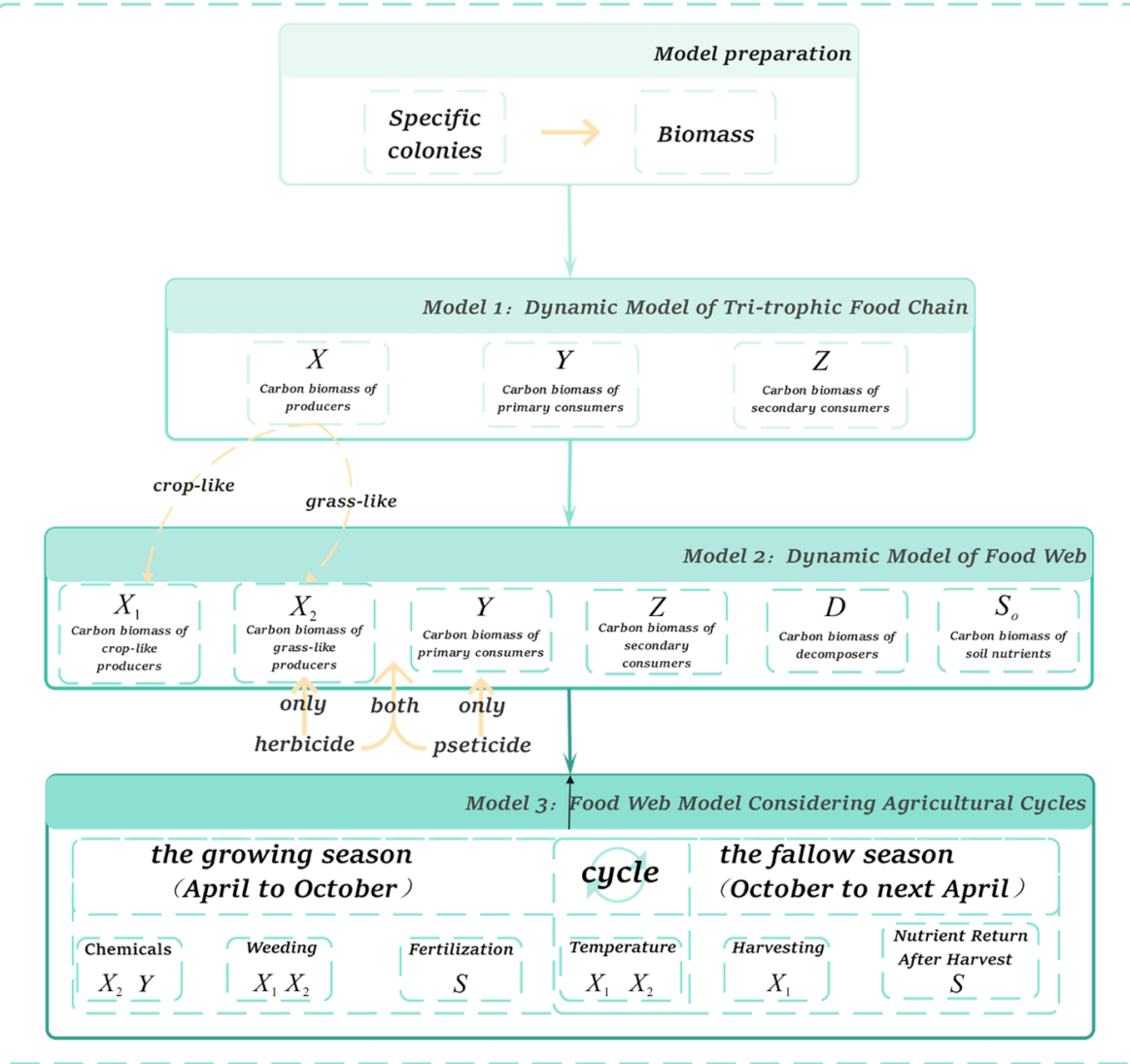


除草剂作用

$$-d_2 X_2$$
$$-d_2 (1 + \alpha e^{-(t-t_0)}) X_2$$

杀虫剂作用

$$\text{引入项 } -\beta e^{-(t-t_0)}$$



模型二

01

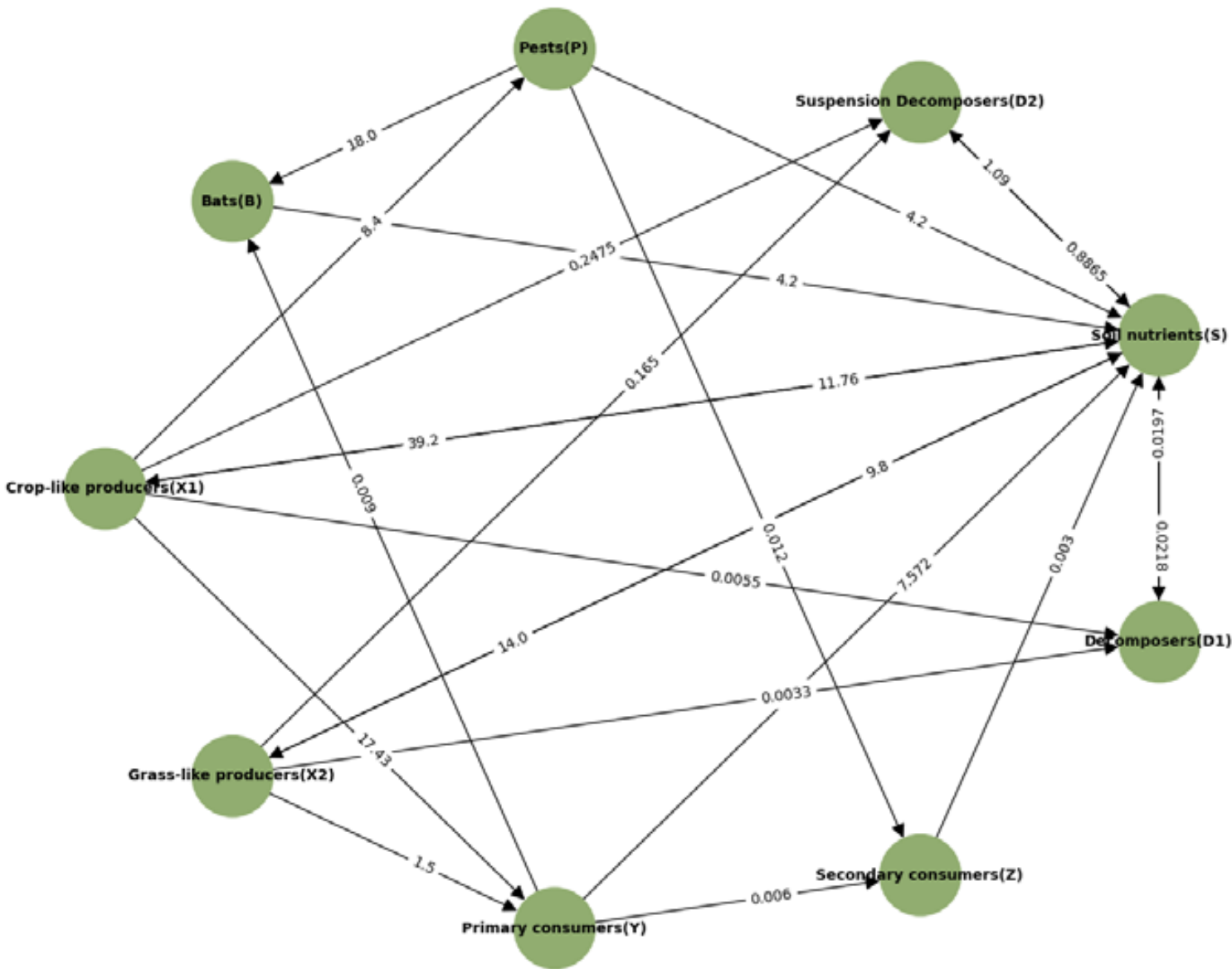
采用有向网络图构建碳流网络，科学描述生态系统中物质与营养的双向流动

02

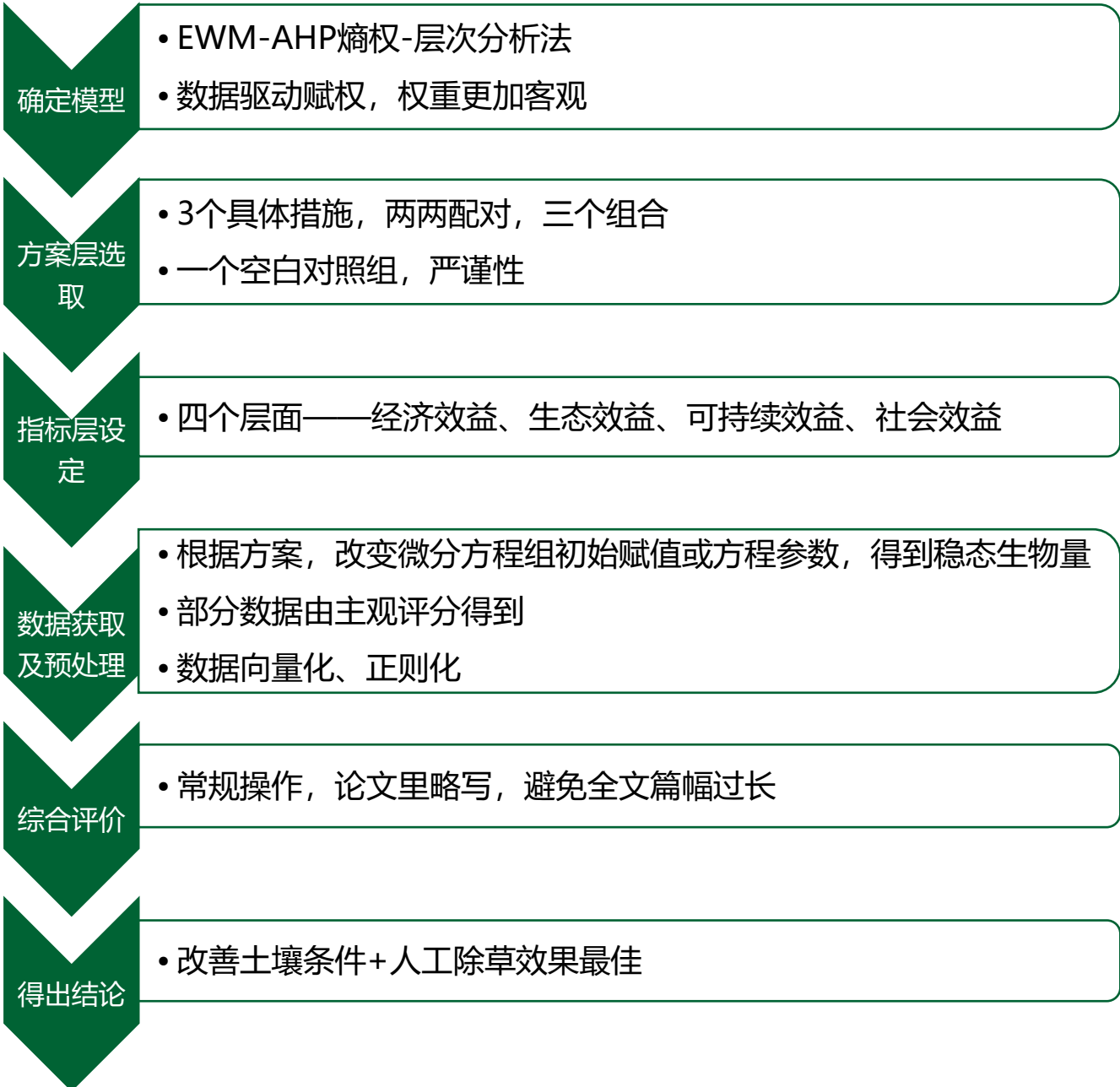
从“动态模拟”到“静态网络结构评估”的视角转换

03

多指标融合评估网络稳定性，“中心性”指标揭示生物学“生态位替代”效应。



模型三



数据集--如何在“零数据”赛题中构建可信基础？

Literature Data (基准参数)

微分方程系数（如捕食率、转化率）无法测量，
直接引用论文的参数初值

Table 4
The determined values of the parameters in the food web dynamic model.

Parameter	Value	Parameter	Value	Parameter	Value
k_0	2	c_{68}	10.115684	a_{57}	0.0800
K	1047.3	I_3	300.863243	a_{58}	0.0800
r_1	85.258286	I_4	1.336978	a_{67}	0.0200
r_2	42.674355	I_5	2.924379	a_{68}	0.2129
d_1	77.402212	I_6	0.801816	e_{13}	0.5800
d_2	41.018143	I_7	3.060976	e_{14}	0.7212
c_{13}	166.066666	I_8	2.324651	e_{24}	0.2000
c_{14}	0.066949	I_P	277.5	e_{25}	0.6000
c_{24}	0.054920	T_r	4.364587	e_{93}	0.8940
c_{25}	0.164510	a_{13}	0.4200	e_{38}	0.2000
c_{93}	0.801161	a_{14}	0.2788	e_{94}	0.8701
c_{38}	6.291640	a_{24}	0.8000	e_{46}	0.8600
c_{94}	0.061361	a_{25}	0.4000	e_{47}	0.9143
c_{46}	0.304755	a_{93}	0.1060	e_{48}	0.2000
c_{47}	0.081738	a_{38}	0.8000	e_{95}	0.8300
c_{48}	0.317309	a_{94}	0.1299	e_{56}	0.0989
c_{95}	0.194329	a_{46}	0.1400	e_{57}	0.9200
c_{56}	3.232027	a_{47}	0.0857	e_{58}	0.9200
c_{57}	1.016859	a_{48}	0.8000	e_{67}	0.9800
c_{58}	4.638478	a_{95}	0.1700	e_{68}	0.7871
c_{67}	2.370563	a_{56}	0.9011		

Zhang, H., et al. (2021). *Ecological Indicators*.

Real-world Data (现实参数)

引入新变量（如“农业周期”、“季节性”），
利用国家级气象数据库（如黑龙江五常市气温），拟合函数驱动模型周期

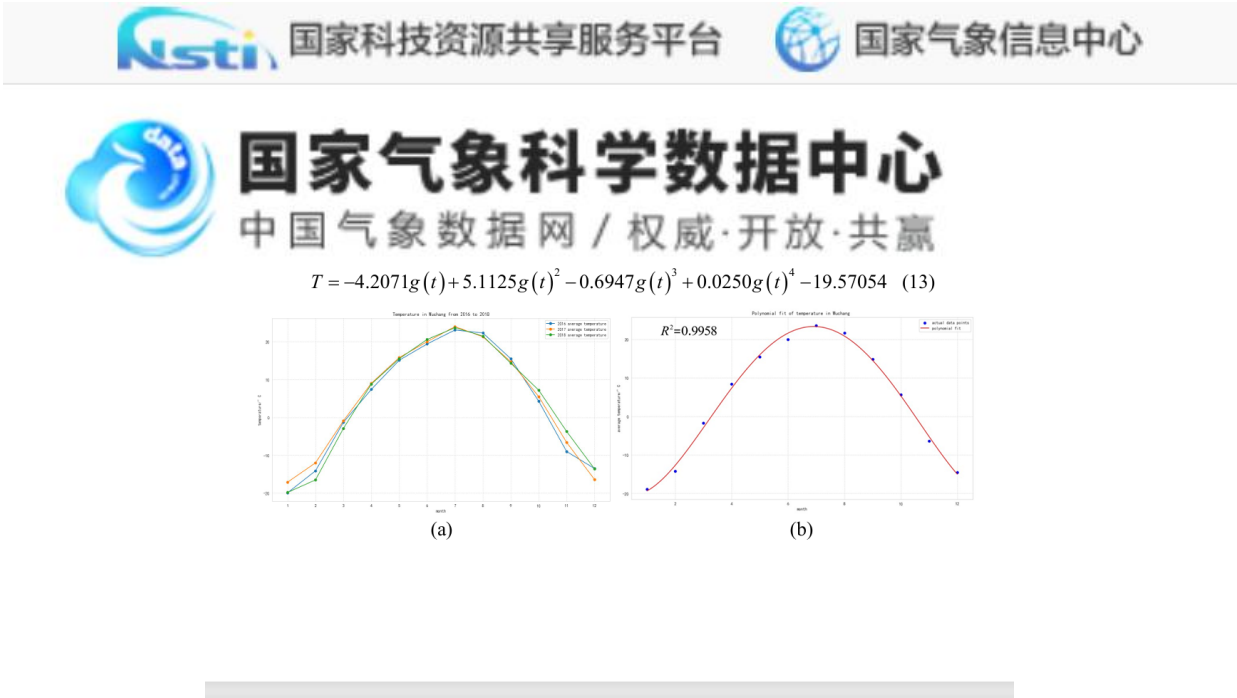
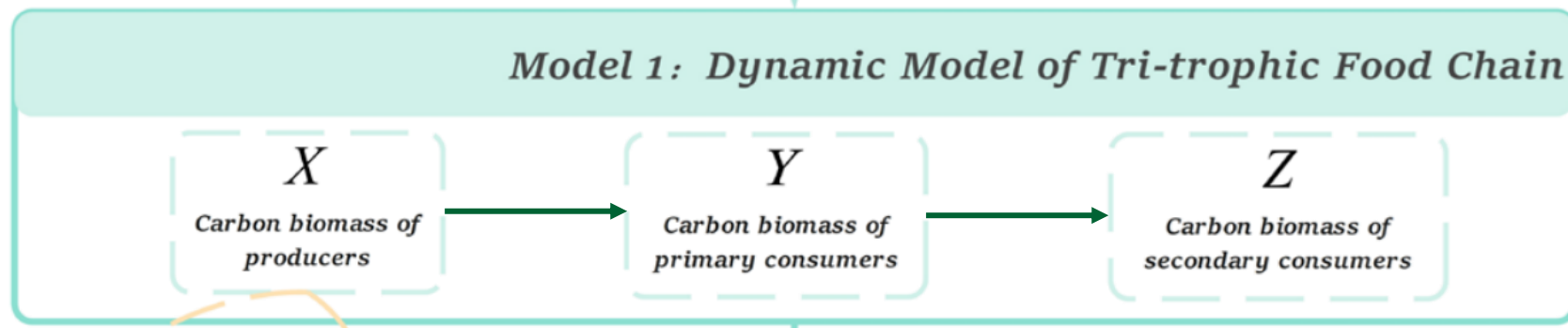


Figure 8: (a) The average temperature data from Wuchang City, Northeast China, from 2016 to 2018 was used for fitting;(b) The resulting quartic polynomial curve

02 | 求解方法

Solution to Problem

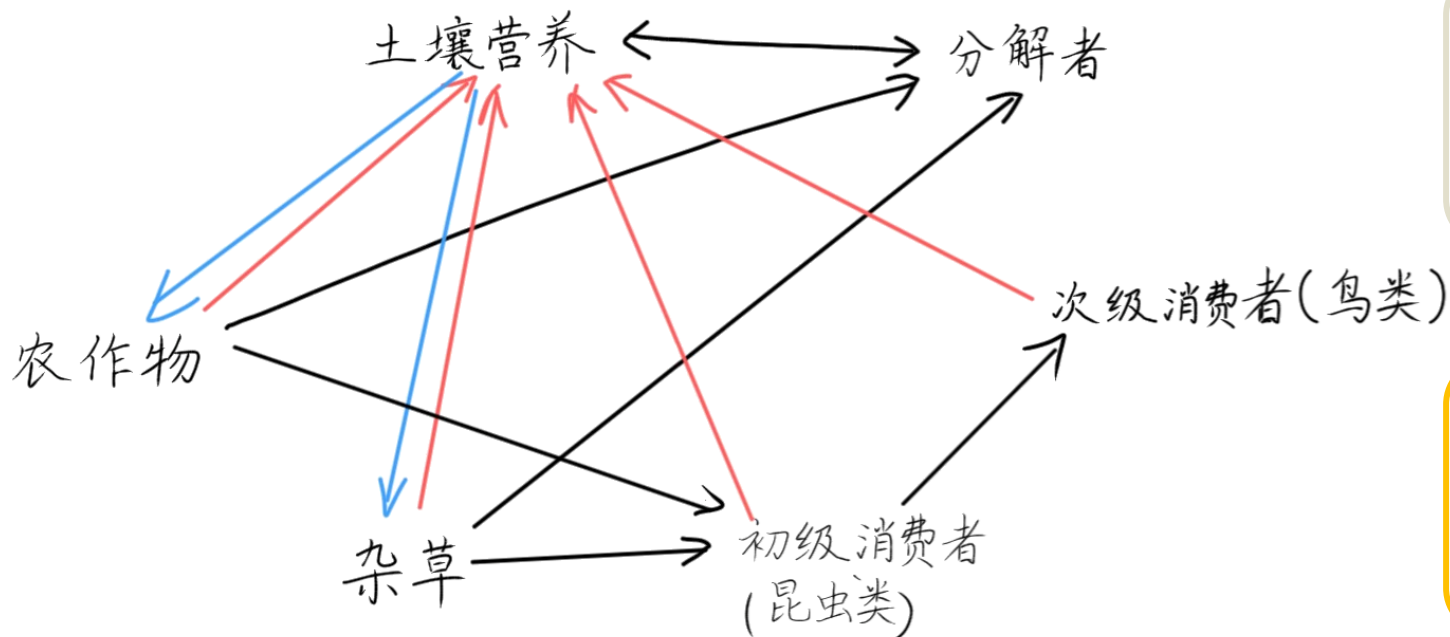
模型1.1——食物链（三层次）



$$\begin{aligned}\frac{dX}{dt} &= X \left(R \left(1 - \frac{X}{K} \right) - \frac{A_1 Y}{B_1 + X} \right) \\ \frac{dY}{dt} &= Y \left(-D_1 + E_1 \frac{A_1 X}{B_1 + X} - \frac{A_2 Z}{B_2 + Y} \right) \\ \frac{dZ}{dt} &= Z \left(-D_2 + E_2 \frac{A_2 Y}{B_2 + Y} \right)\end{aligned}$$

种间竞争的实质是两种或更多种生物共同利用同一**有限资源**(如食物、空间、水分等)时, 因**资源有限**而产生的相互抑制、相互排斥的相互作用。Lotka-Volterra 种间竞争模型字母所表含义如下:

R 是食饵种群的内禀增长率, K 是食饵种群的环境承载力, A_i 分别代表中间捕食者和顶级捕食者的最大捕食率, B_i 为半饱和常数, D_i 分别是中间捕食者和顶级捕食者的死亡率, E_i 分别表示中间捕食者对食饵捕食的转化系数和顶级捕食者对中间捕食者捕食的转化系数, T 表示时间。初始条件为 $P(0) \geq 0, Z(0) \geq 0, F(0) \geq 0$ 。



同模型一的变化:

- 1.生产者分为农作物和杂草两类
- 2.引入分解者和土壤

红色箭头: 1.生产者尸体, 2.消费者粪便
蓝色箭头: 生产者汲取土壤养分

before

$$\begin{aligned}\frac{dX}{dt} &= X \left(R \left(1 - \frac{X}{K} \right) - \frac{A_1 Y}{B_1 + X} \right) \\ \frac{dY}{dt} &= Y \left(-D_1 + E_1 \frac{A_1 X}{B_1 + X} - \frac{A_2 Z}{B_2 + Y} \right) \\ \frac{dZ}{dt} &= Z \left(-D_2 + E_2 \frac{A_2 Y}{B_2 + Y} \right)\end{aligned}$$

after

$$\begin{aligned}\frac{dX_1}{dt} &= r_1 X_1 \left(1 - \frac{X_1 + X_2}{K} \right) - c_{13} X_1 Y - c_{15} X_1 D + a_{61} c_{61} S X_1 - d_1 X_1 \\ \frac{dX_2}{dt} &= r_2 X_2 \left(1 - \frac{X_1 + X_2}{K} \right) - c_{23} X_2 Y - c_{25} X_2 D + a_{62} c_{62} S X_2 - d_2 X_2 \\ \frac{dY}{dt} &= a_{13} c_{13} X_1 Y + a_{23} c_{23} X_2 Y - c_{34} Y Z - I_3 Y \\ \frac{dZ}{dt} &= a_{34} c_{34} Y Z - I_4 Z \\ \frac{dD}{dt} &= a_{15} c_{15} X_1 D + a_{25} c_{25} X_2 D + a_{65} c_{65} S D - I_5 D \\ \frac{dS}{dt} &= I_S + d_1 X_1 + d_2 X_2 + e_{13} c_{13} X_1 Y + e_{23} c_{23} X_2 Y + e_{34} c_{34} Y Z + e_{15} c_{15} X_1 D + e_{25} c_{25} X_2 D \\ &\quad + e_{61} c_{61} S X_1 - c_{61} S X_1 + e_{62} c_{62} S X_2 - c_{62} S X_2 + e_{65} c_{65} S D - c_{65} S D - T_r S\end{aligned}$$

农作物: $\frac{dX_1}{dt} = r_1 X_1 \left(1 - \frac{X_1 + X_2}{K} \right) - c_{13} X_1 Y - c_{15} X_1 D + a_{61} c_{61} S X_1 - d_1 X_1$

杂草: $\frac{dX_2}{dt} = r_2 X_2 \left(1 - \frac{X_1 + X_2}{K} \right) - c_{23} X_2 Y - c_{25} X_2 D + a_{62} c_{62} S X_2 - d_2 X_2$

初级消费者: $\frac{dY}{dt} = a_{13} c_{13} X_1 Y + a_{23} c_{23} X_2 Y - c_{34} Y Z - I_3 Y$

次级消费者: $\frac{dZ}{dt} = a_{34} c_{34} Y Z - I_4 Z$

土壤分解者: $\frac{dD}{dt} = a_{15} c_{15} X_1 D + a_{25} c_{25} X_2 D + a_{65} c_{65} S D - I_5 D$

$\frac{dS}{dt} = I_s + d_1 X_1 + d_2 X_2 + e_{13} c_{13} X_1 Y + e_{23} c_{23} X_2 Y + e_{34} c_{34} Y Z + e_{15} c_{15} X_1 D + e_{25} c_{25} X_2 D + e_{61} c_{61} S X_1 - c_{61} S X_1 + e_{62} c_{62} S X_2 - c_{62} S X_2 + e_{65} c_{65} S D - c_{65} S D - T_r S$

I_i : 消费者 i 的呼吸速率

d_i : 死亡率

系数 a_{ij} : j 消费 i 时的同化效率

c_{ij} : j 消费 i 时的最大消耗速率 ($c_{ij} = \frac{CF_{ij}}{Bm_i \times Bm_j}$)

e_{ij} : j 对 i 的排泄率 ($e_{ij} = 1 - a_{ij}$)

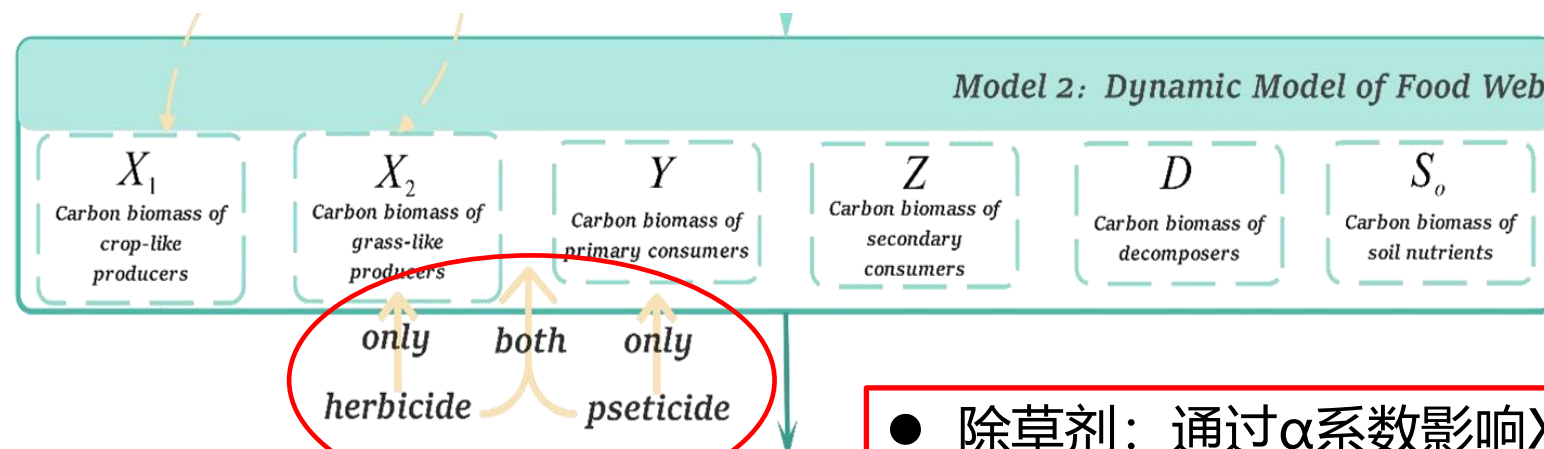
大写字母 X_1, X_2, S 等: 生物量 biomass

I_s : 森林残余物的悬浮有机物输入量

T_r : 总输出率

森林相关

模型1.2.2——考虑除草剂、杀虫剂

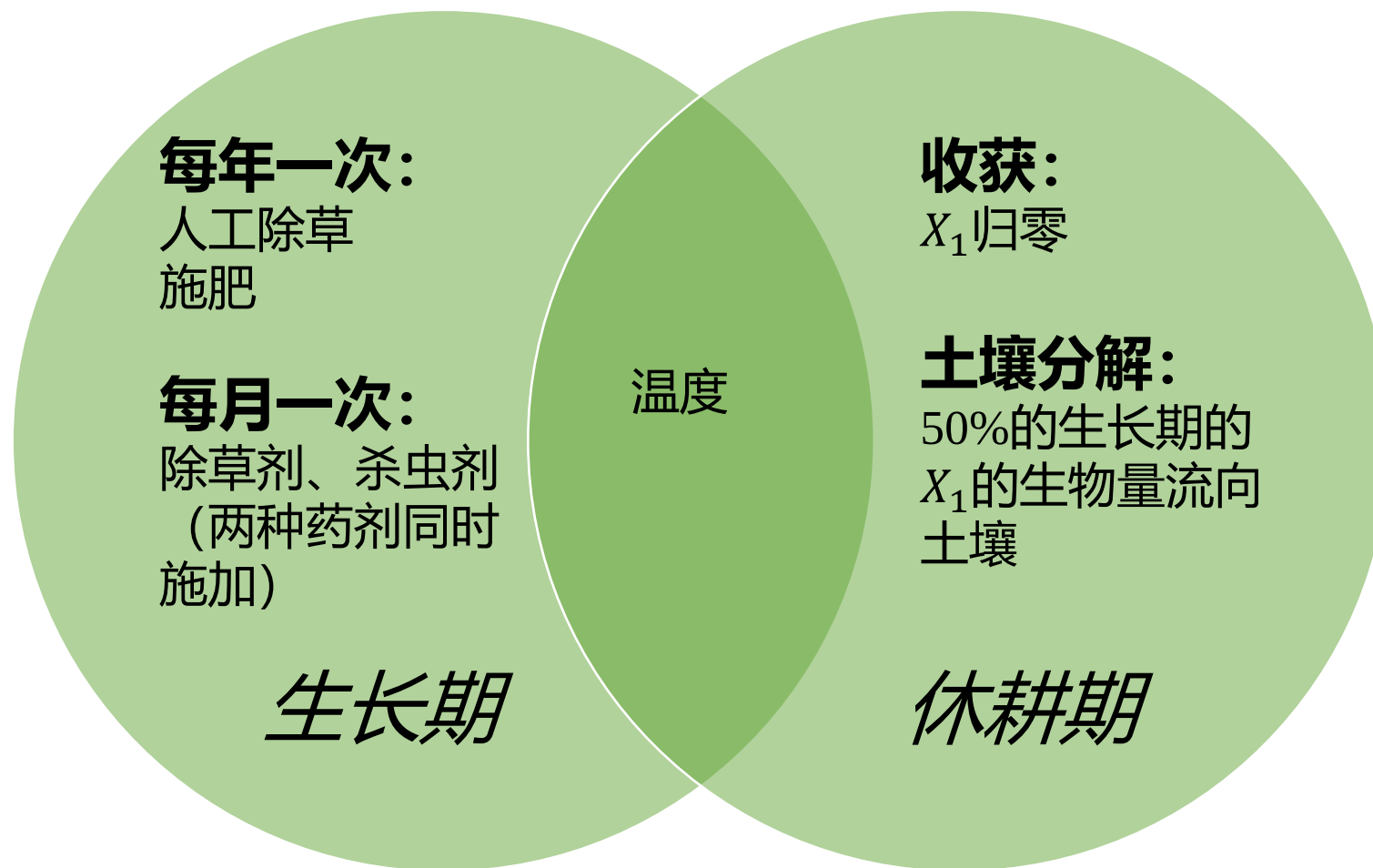


- 除草剂：通过 α 系数影响 X_2 ，作用效果并非立竿见影，较为缓慢
- 杀虫剂：通过 β 系数直接作用于初级消费者 Y ，效果立竿见影

$$\frac{dX_2}{dt} = r_2 X_2 \left(1 - \frac{X_1 + X_2}{K} \right) - c_{23} X_2 Y - c_{25} X_2 D + a_{62} c_{62} S X_2 - d_2 X_2 \longrightarrow \frac{dX_2}{dt} = r_2 X_2 \left(1 - \frac{X_1 + X_2}{K} \right) - c_{23} X_2 Y - c_{25} X_2 D + a_{62} c_{62} S X_2 - d_2 (1 + \alpha e^{-(t-t_0)}) X_2$$

$$\frac{dY}{dt} = a_{13} c_{13} X_1 Y + a_{23} c_{23} X_2 Y - c_{34} Y Z - I_3 Y \longrightarrow \frac{dY}{dt} = a_{13} c_{13} X_1 Y + a_{23} c_{23} X_2 Y - c_{34} Y Z - I_3 Y - \beta e^{-(t-t_0)}$$

模型1.3——考虑农业周期



生长期

施肥作用

土壤条件每个生长期初始增加固定值， S_p 是上一农业周期土壤条件期末值

$$S = S_p + 10$$

人工除草作用

初始值 $X_1 : X_2 = 10 : 1$

除草剂、杀虫剂作用

同前，变化之处在于作用的周期性，周期为h

共同作用

温度作用

改变X1和X2生长系数，影响X1和X2的生长速率

$$\frac{dX_1}{dt} = [-0.005(T - 16)^2 + 2r_1]X_1 \left(1 - \frac{X_1 + X_2}{K}\right) + \dots$$
$$\frac{dX_2}{dt} = [-0.005(T - 16)^2 + 2r_2]X_2 \left(1 - \frac{X_1 + X_2}{K}\right) + \dots$$

休耕期

收获

X_1 清零，微分方程组中含 X_1 的项清除

分解

50%的生长期的 X_1 的生物量流向土壤

生长期	休耕期
$\frac{dX_1}{dt} = \left[-0.005(T-16)^2 + 2r_1\right] X_1 \left(1 - \frac{X_1 + X_2}{K}\right) + \dots - d_1(1 + \alpha e^{-k(t)}) X_1$$\frac{dX_2}{dt} = \left[-0.005(T-16)^2 + 2r_2\right] X_2 \left(1 - \frac{X_1 + X_2}{K}\right) + \dots - d_2(1 + \alpha e^{-k(t)}) X_2$	$\frac{dX_2}{dt} = \left[-0.005(T-16)^2 + 2r_2\right] X_2 \left(1 - \frac{X_1 + X_2}{K}\right) + \dots - d_2 X_2$
$\frac{dY}{dt} = a_{13} c_{13} X_1 Y + a_{23} c_{23} X_2 Y - c_{34} YZ - I_3 Y$	$\frac{dY}{dt} = a_{23} c_{23} X_2 Y - c_{34} YZ - I_3 Y$
$\frac{dD}{dt} = a_{15} c_{15} X_1 D + a_{25} c_{25} X_2 D + a_{65} c_{65} SD - I_5 D$	$\frac{dD}{dt} = a_{25} c_{25} X_2 D + a_{65} c_{65} SD - I_5 D$
$\frac{dS}{dt} = I_S + d_1 X_1 + d_2 X_2 + e_{13} c_{13} X_1 Y + e_{23} c_{23} X_2 Y + e_{34} c_{34} YZ + e_{15} c_{15} X_1 D + e_{25} c_{25} X_2 D$$- e_{61} c_{61} S X_1 - c_{61} S X_1 + e_{62} c_{62} S X_2 - c_{62} S X_2 + e_{65} c_{65} S D - c_{65} S D - T_r S$	$\frac{dS}{dt} = I_S + d_2 X_2 + e_{23} c_{23} X_2 Y + e_{34} c_{34} YZ + e_{25} c_{25} X_2 D$$+ e_{62} c_{62} S X_2 - c_{62} S X_2 + e_{65} c_{65} S D - c_{65} S D - T_r S + \frac{1}{2} X_p$

Problem 2: Ecosystem Assessment

灵感来源

社会计算课程

社会网络相关的知识 可以迁移到 食物网中

1-网络指标 => 评估食物网

2-中心性 => 刻画生物重要性

图论：离散数学

下午 5:05 12月3日周三

4. 信息流和介数 betweenness

1) 介数 = 信息流点数 有大量并不冗余的最大路径通过

2) 跨越结构洞的, 介数高

5. 中心性 1) 接近中心性: 一个节点与其它节点靠近, 在信息传播中越不依赖其它节点, 与其它节点距离之和越小, 该节点越重要

2) 特征向量中心性: 一个节点重要性不仅取决于邻居节点数量, 也取决于与该节点相连的节点的重要性

PageRank



Gephi

<https://gephi.org>

Gephi - The Open Graph Viz Platform

统计 设置

网络概述

平均度 运行

平均加权度 运行

网络直径 5 运行

图密度 运行

点击次数 运行

模块化 0.557 运行

PageRank 运行

连接组件 运行

节点概述

平均聚类系数 运行

特征向量中心度 运行

边概述

平均路径长度 2.4 运行

统计名称

点击进行统计

运行后点击问号可以查看报告

Problem 2: Ecosystem Assessment

1. 食物网评估

1-Food Web Connectivity 连通性：结构上

$$C = \frac{M}{N \cdot N}, L_a = \frac{\sum L_{ij}}{M}$$

2-Maximum Eigenvalue Method

最大特征向量：功能上

$$\lambda_{\max} = \max(\operatorname{Re}(\lambda_i)) \quad (24)$$

Where, $\lambda_{\max}$ represents the maximum real eigenvalue; λ_i represents is the i -th eigenvalue of the adjacency matrix; $\operatorname{Re}(\lambda)$ represents denotes the real part of the eigenvalue;

3- Shannon-Wiener index

$$H = -\sum_{i=1}^S p_i \ln p_i \quad H = -\sum_{i,j} CF_{ij} \ln CF_{ij}$$

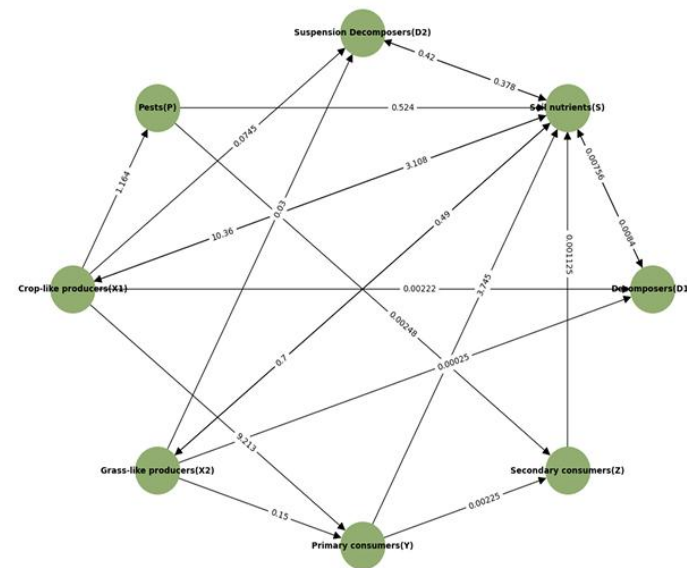


Figure 10: food web after introducing new species result

$$CF_{jk} = c_{jk} \cdot BM_j \cdot BM_k$$

Problem 2: Ecosystem Assessment

2•物种重要性评估

1-Betweenness Centrality 介数中心性

$$C_B(i) = \sum_{s,t \neq i} \frac{\sigma_{st}(i)}{\sigma_{st}}$$

2-Closeness Centrality 接近中心性

$$C_C(i) = \frac{1}{\sum_{j \neq i} d(i, j)}$$

3-Eigenvector Centrality 特征向量中心性

$$C_E(i) = \frac{1}{\lambda} \sum_j CF_{ij} C_E(j)$$

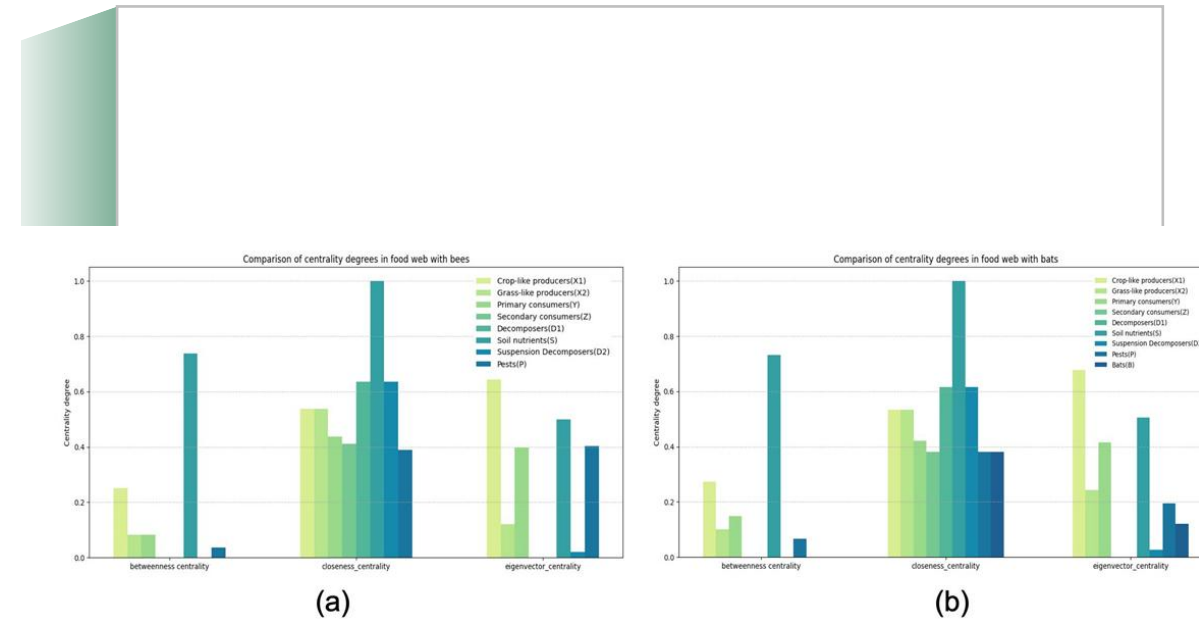
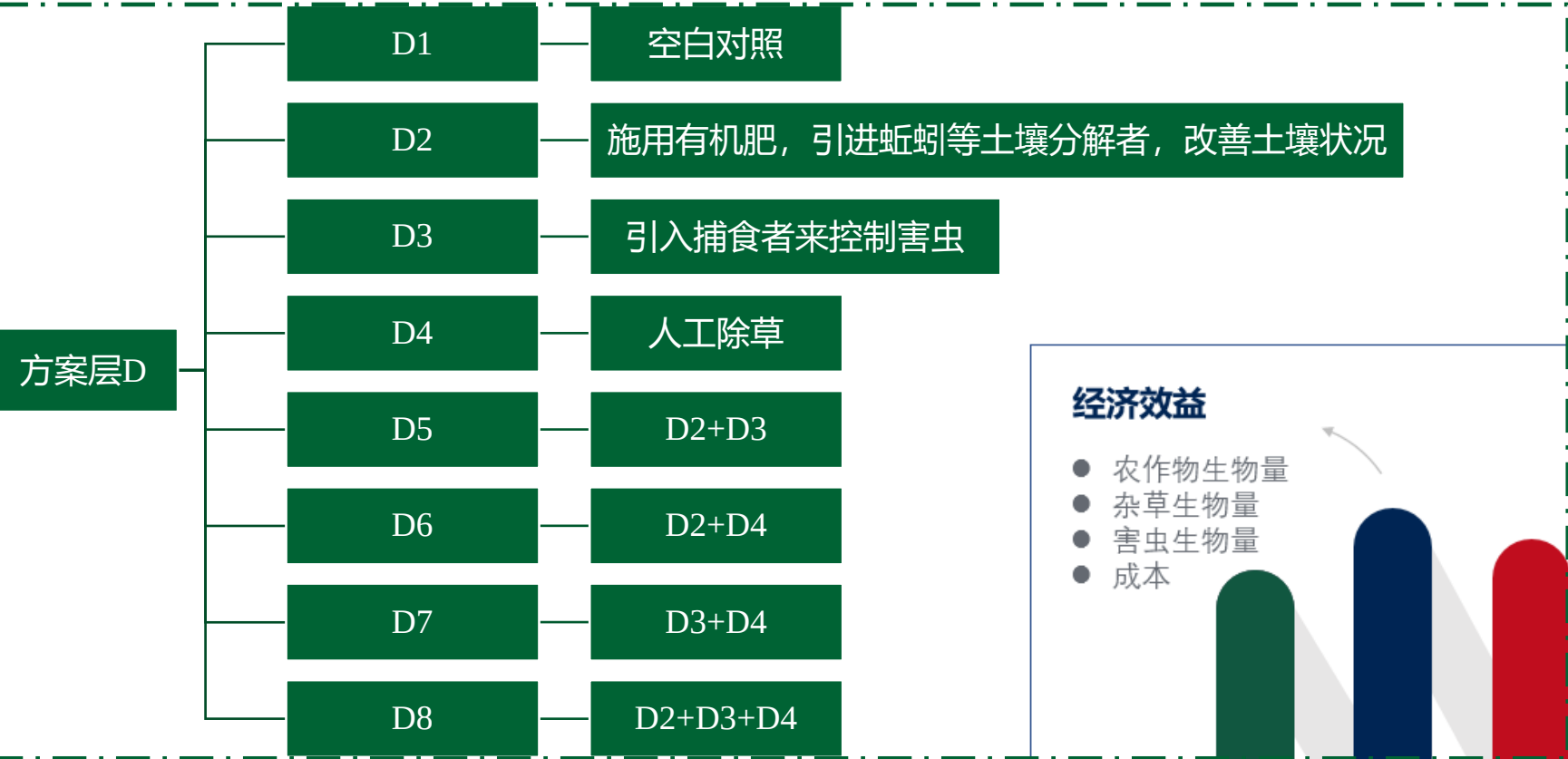
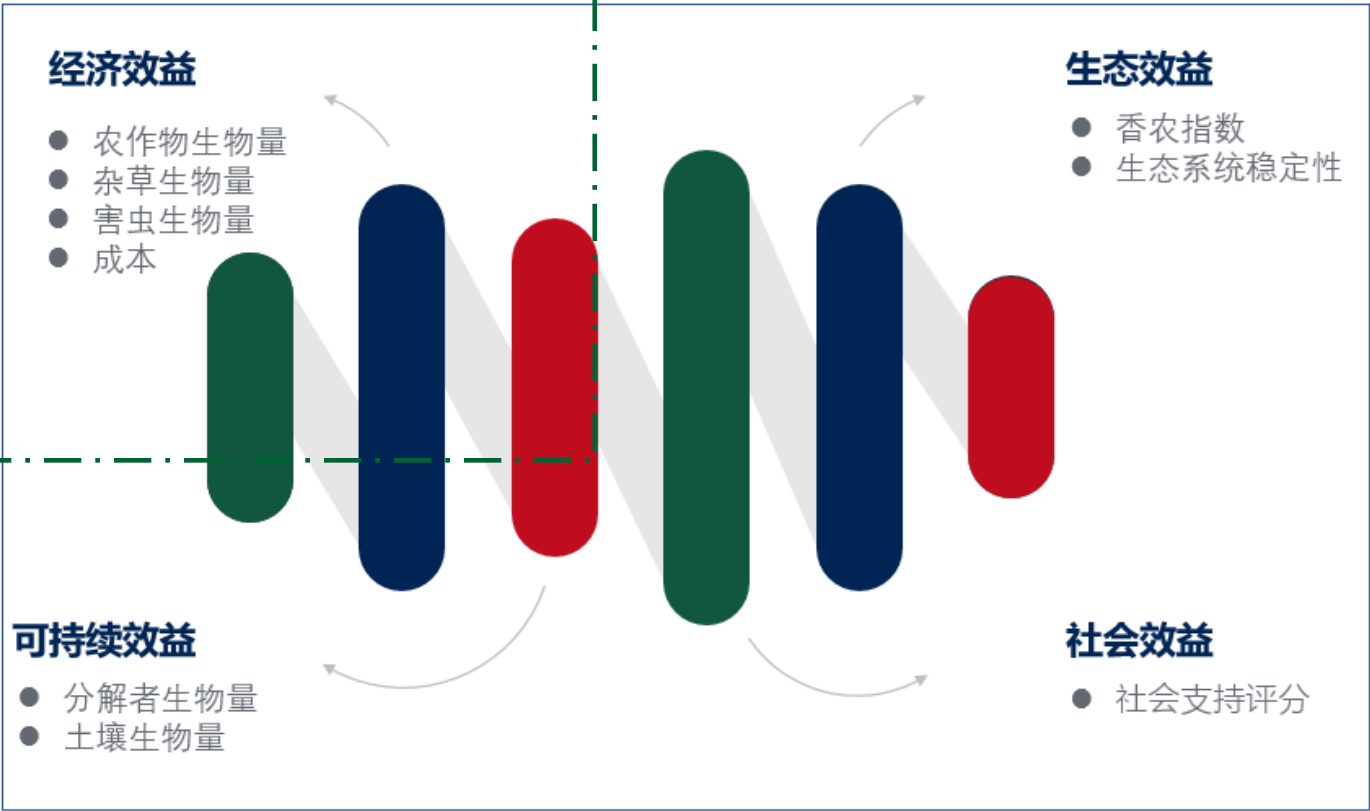


Figure 13: (a) Comparison of centrality in degrees in food web with bees; (b) Comparison of centrality in degrees in food web with bats



熵权-层次分析法
实现农业评价

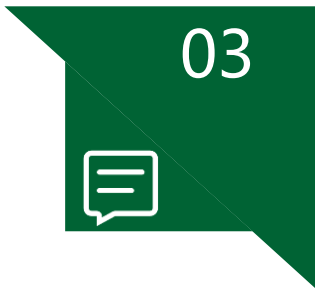
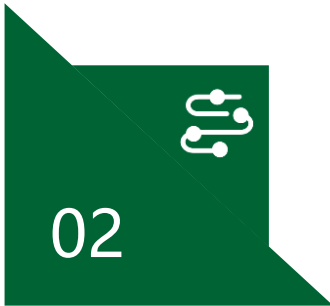
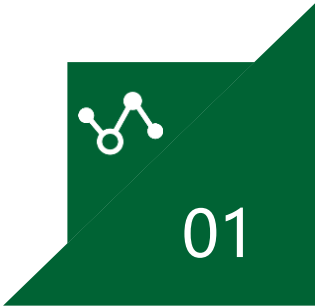


EWM-AHP法

既结合指标数据计算权重，又融入专家决策、主观评分。

社会支持评分

将“政策支持”纳入模型。当农户采用的绿色措施越多，产品在广告、营销、政府有机认证等方面的优势越大，评级就越高。

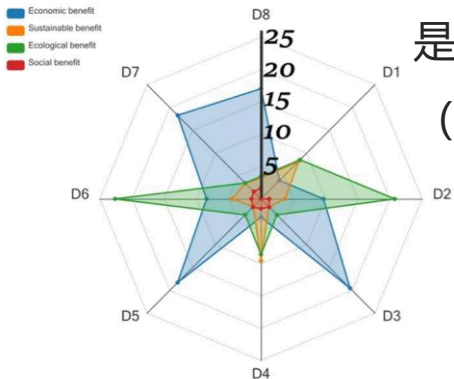


成本核算评分

主观评分指标，主要考虑资金成本、人力成本和时间成本。人工除草会显著增加劳动力和时间成本；施肥生效周期长，时间成本也会增加。

主要结论

方案D6（改善土壤条件 + 定期除草）是最优解。其在维持生态系统稳定性（权重超30%）方面表现最佳。



Sensitivity Analysis

是什么？

1-robustness 模型是否稳健？

参数改变，模型、结果是否依然适用

=>适用范围足够广

Eg. 右端项 b （资源）变化多少范围内，最优基不变？

目标函数系数 c 变化多少范围内，最优解不变？

添加新的约束/变量会怎样？

2-参数取值是否合理？

Eg. 我们论证了农药浓度选取的合理性

3-哪些参数最重要？

线性规划的对偶问题

Eg. shadow price（对偶变量）：表现资源稀缺性

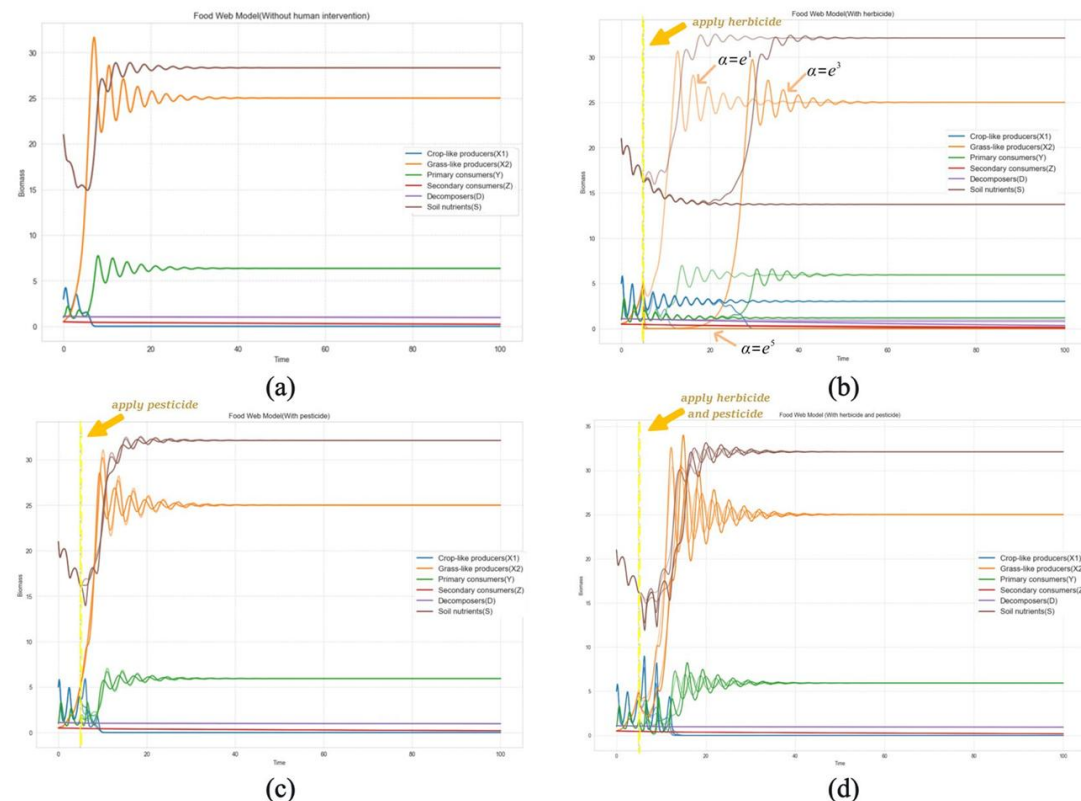


Figure 7: (a) Without Human Intervention; (b) With Herbicide; (c) With Pesticide; (d) With Herbicide and Pesticide

Sensitivity Analysis



Bifurcation diagrams

灵感来源

Employing dynamic model to study food web stability and possible regime shifts in Somme Bay

选择alpha 除草剂剂量

如何解释很重要

自圆其说

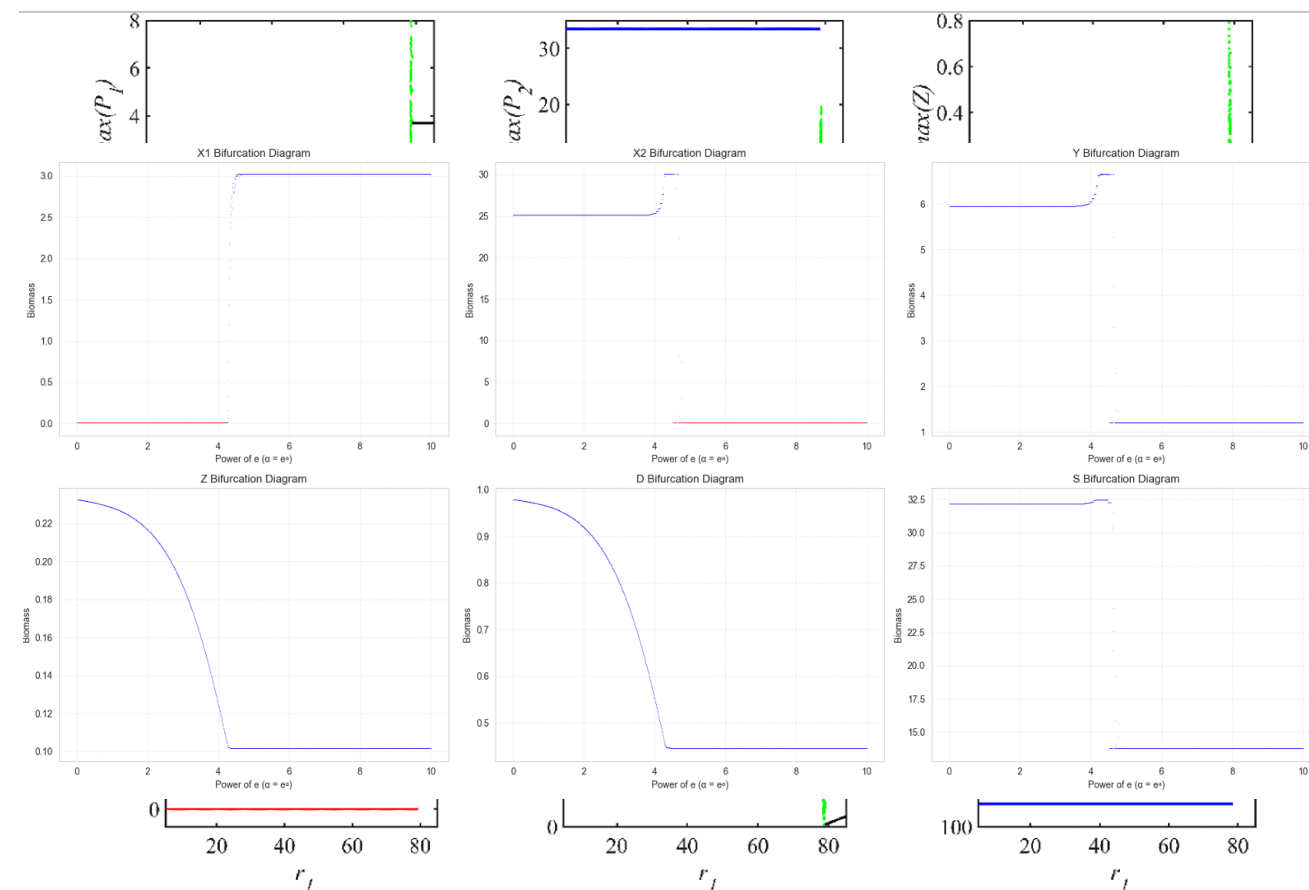
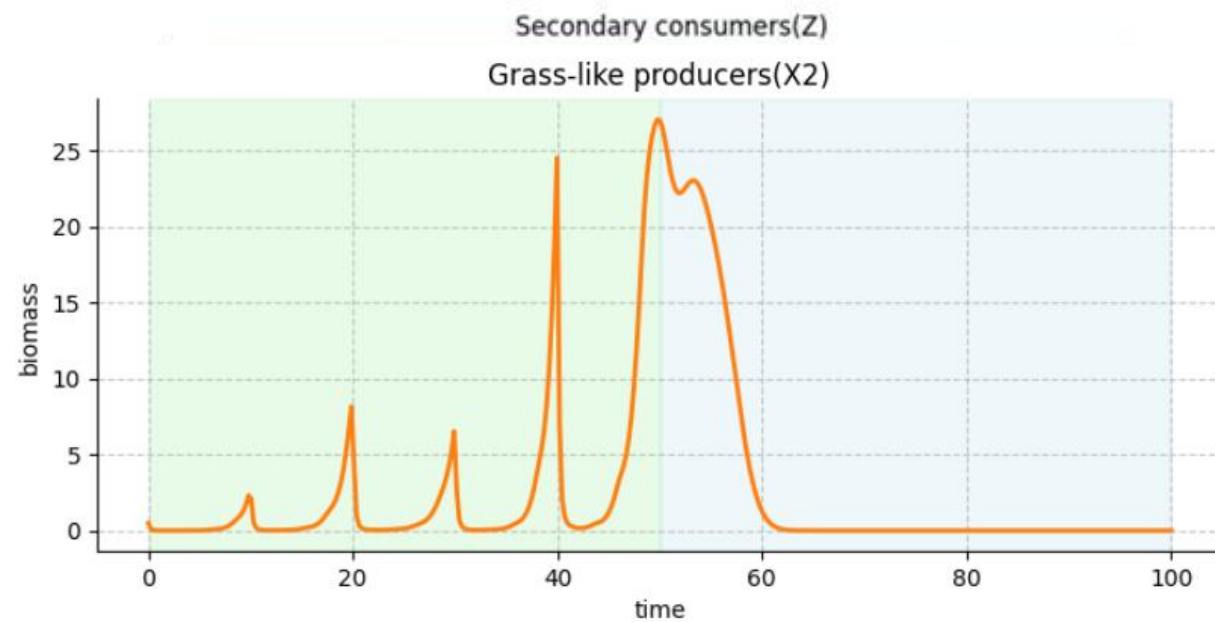
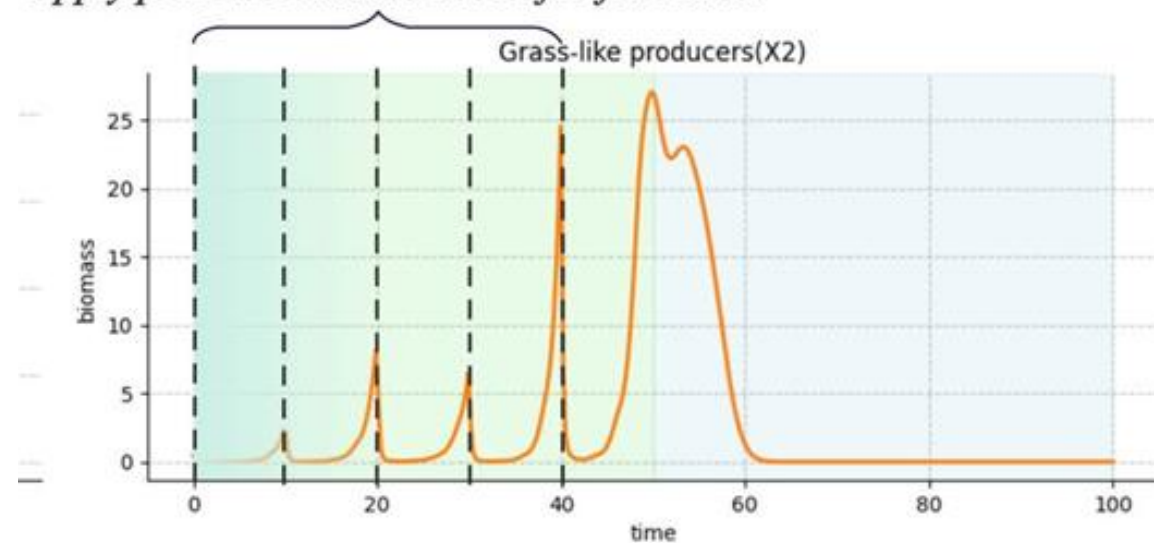


Fig. 2. Bifurcation diagrams of the nine taxonomic groups when r_1 value ranges in [5.27, 85.5], demonstrating the shift of dynamical states with the parameter variation. Four different dynamical states are present: black for equilibrium, blue for periodic oscillation, green for chaotic oscillation, and red for extinction. Corresponding to one given value of r_1 , the biomass of each taxonomic group would exhibit one of the four dynamical states. For the periodic and chaotic oscillations, the maxima for the biomass of P_1 , P_2 , Z , S , D , P , B , F , P_m , respectively, is utilized in plotting the graphs. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

编程

python & excel

1-数值模拟、绘图

Apply pesticide and herbicide for five times

编程

python & excel

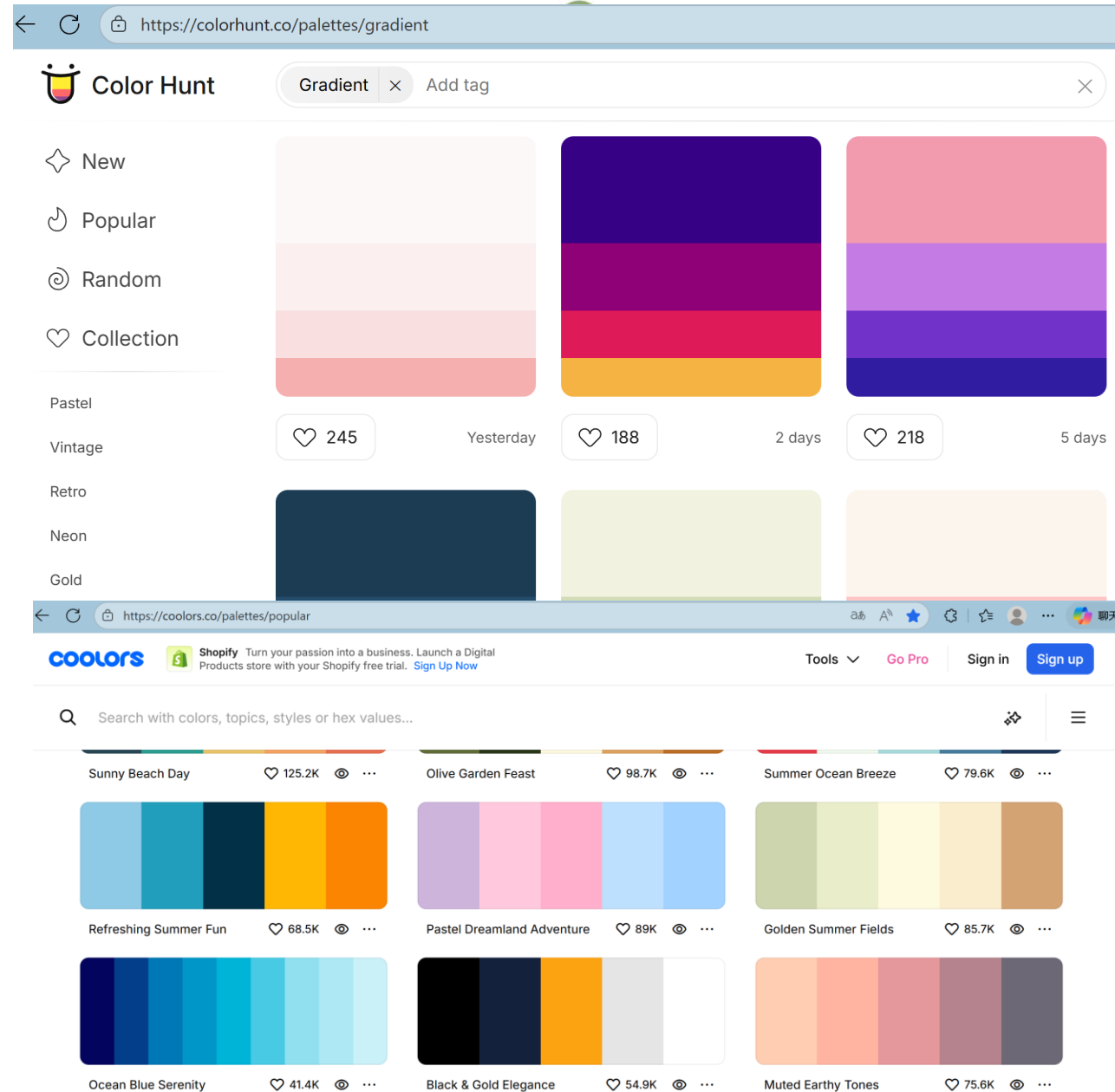
1-数值模拟、绘图

2-图论部分

nippon-color : <https://ssshooter.github.io/nippon-color/#/>

color hunt : <https://colorhunt.co/>

coolers : <https://coolers.co/palettes/popular>



编程

python & excel

1-数值模拟、绘图

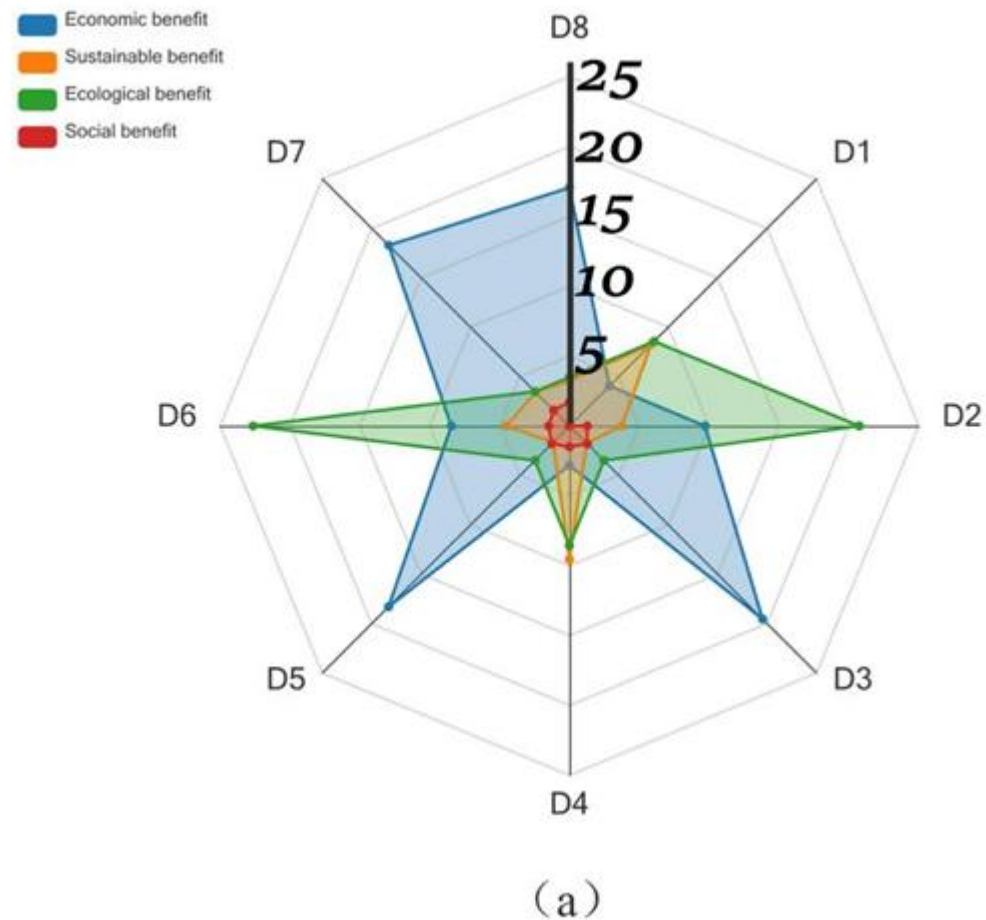
2-图论部分

nippon-color : <https://ssshooter.github.io/nippon-color/#/>

color hunt : <https://colorhunt.co/>

coolers : <https://coolers.co/palettes/popular>

3-excel EasyShu



论文写作 —— 逻辑和语言

赛前准备：

- 基本模型知识

不需要融会贯通，但至少理解运用该模型解决（简化）问题的逻辑

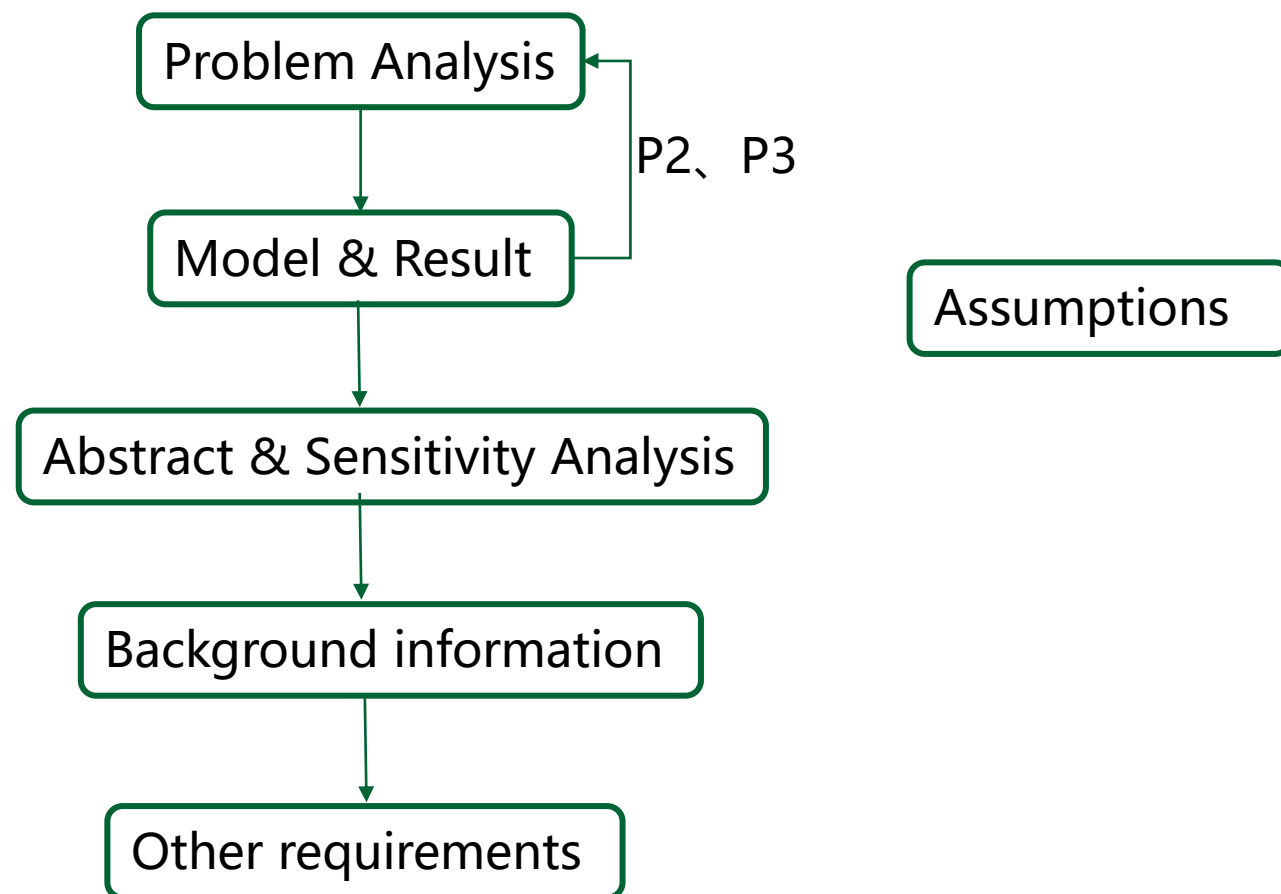
- 广读历年O奖论文

学习书写逻辑，赛中无法下笔时可以再反复琢磨

- 英语能力（巧用gpt）

学会使用gpt，除了润色还能教会你很多别的东西

写作顺序：

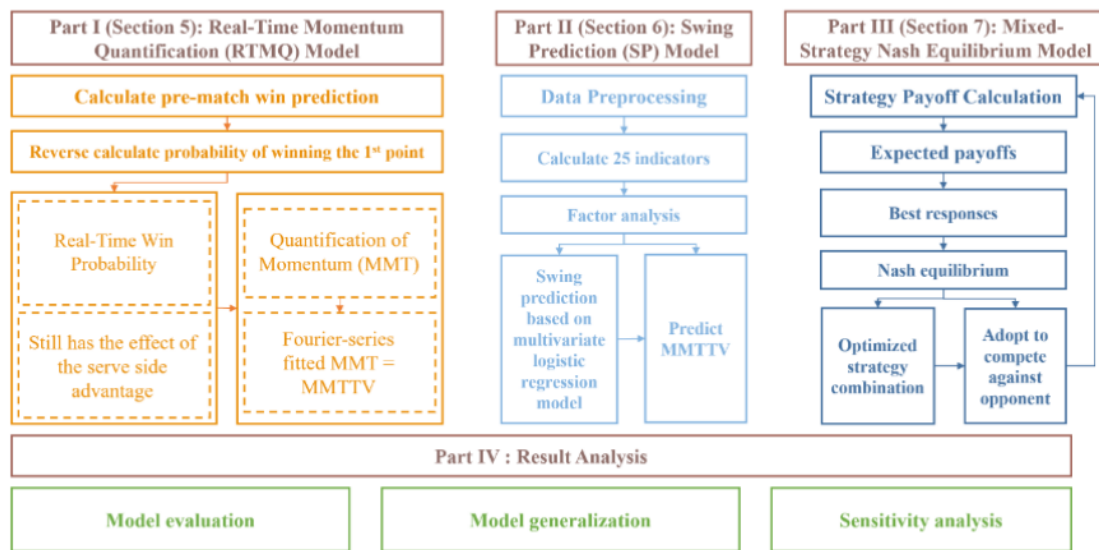


论文写作 —— 画图与格式

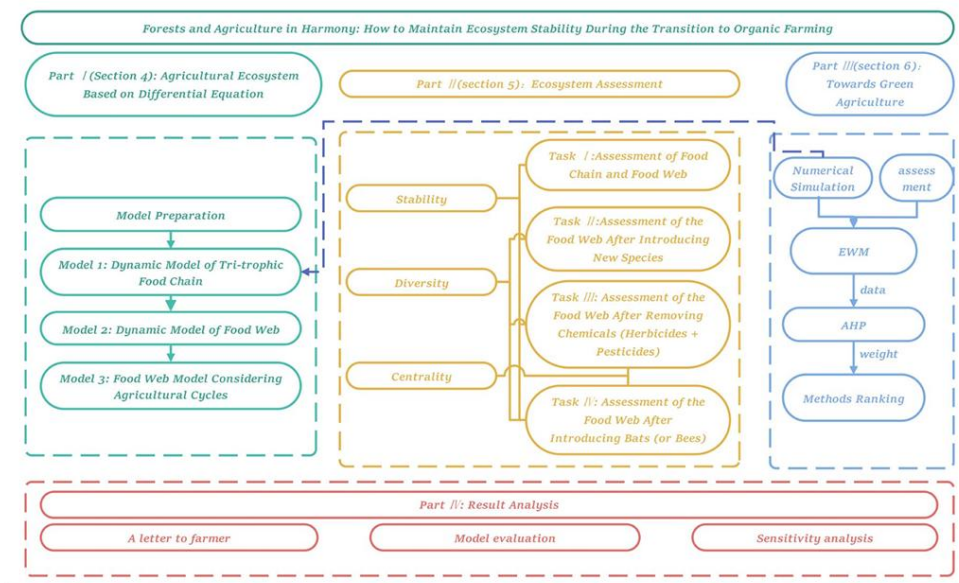
画图软件准备：

亿图图示、PPT、Excel（插件Easyshu）

- 提前大量练习，模仿O奖论文的图
- 必画图Workflow一定要提前准备好自己的一个模板
- 细节：风格统一



24年O奖论文



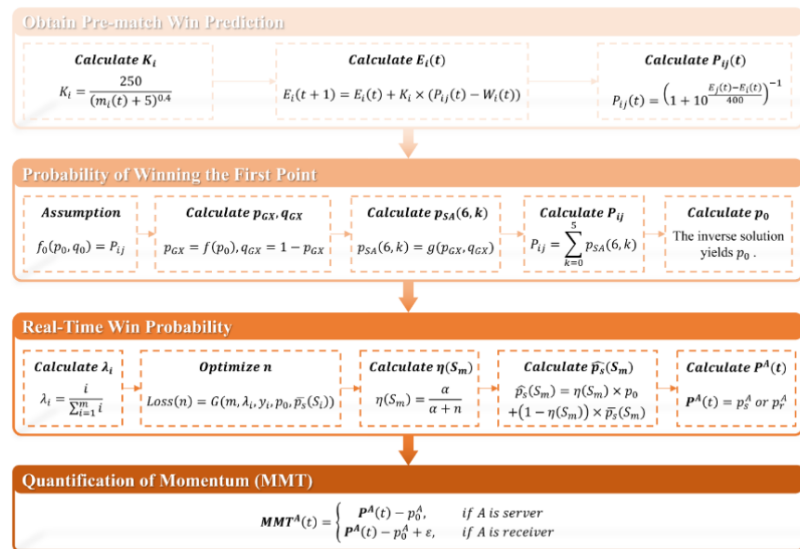
我们的论文

论文写作 —— 画图与格式

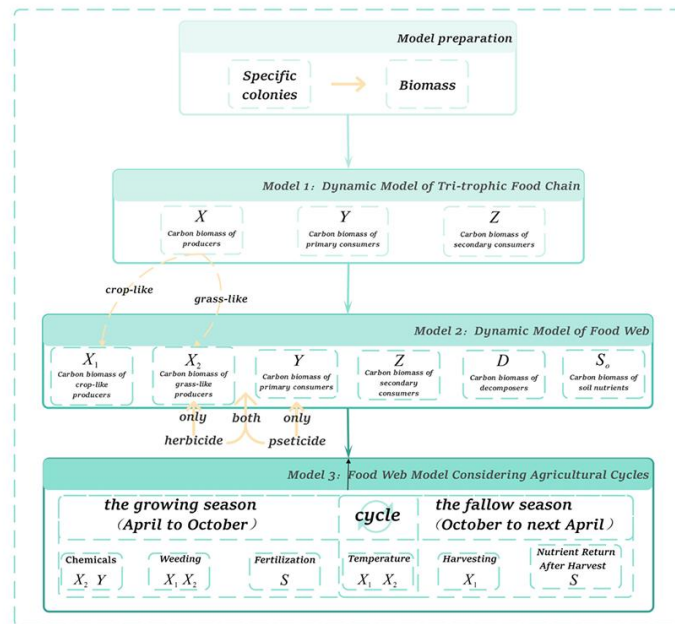
画图软件准备：

亿图图示、PPT、Excel（插件Easyshu）

- 提前大量练习，模仿O奖论文的图
- 必画图Workflow一定要提前准备好自己的一个模板
- 细节：风格统一



24年O奖论文



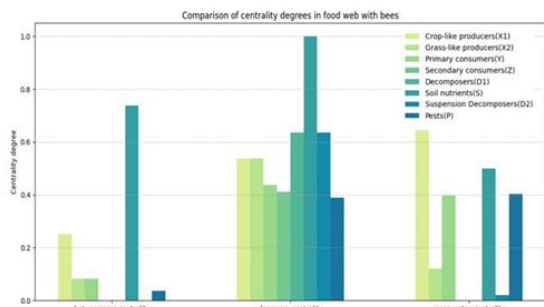
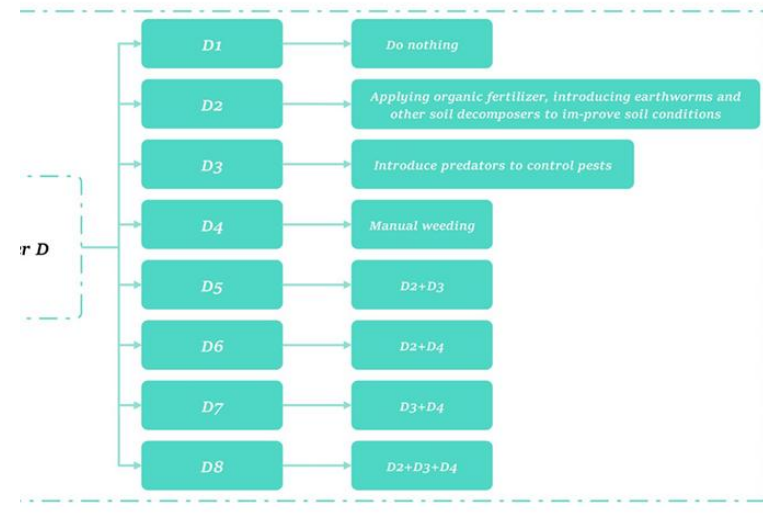
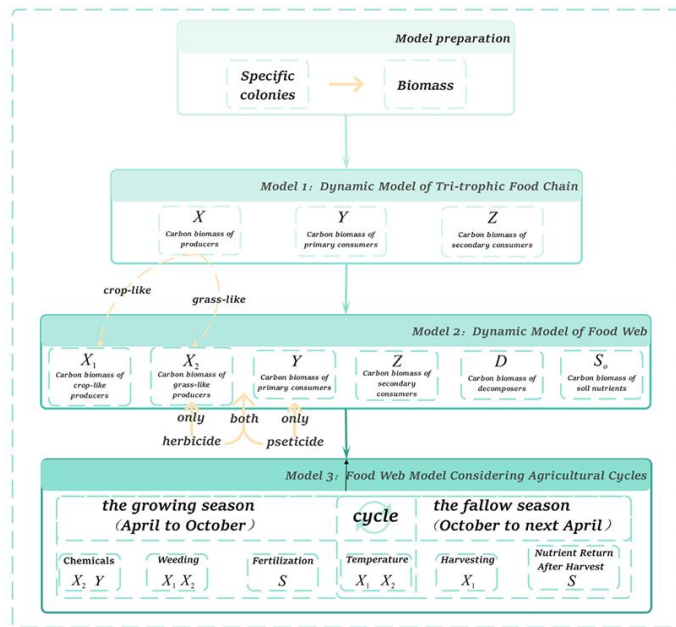
我们的论文

论文写作 —— 画图与格式

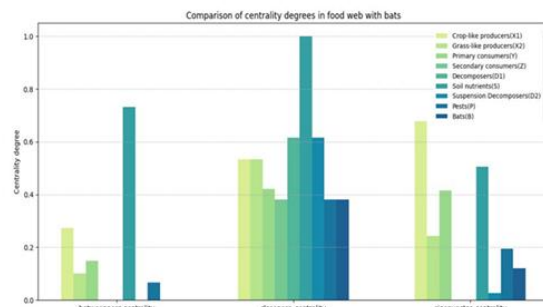
画图软件准备：

亿图图示、PPT、Excel（插件Easyshu）

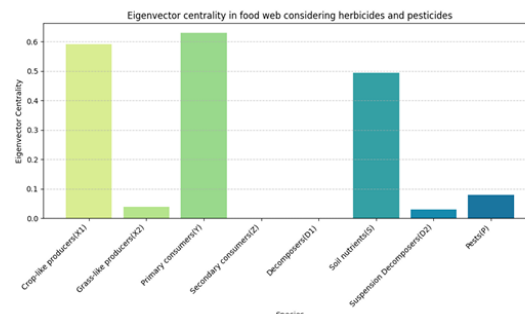
- 提前大量练习，模仿O奖论文的图
- 必画图Workflow一定要提前准备好自己的一个模板
- 细节：风格统一



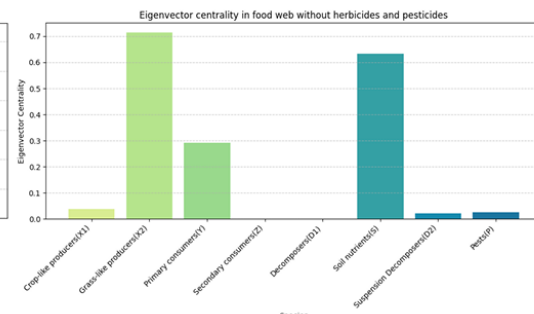
(a)



(b)



(a)



(b)

论文写作 —— 画图与格式

文档准备：

Word模板

- 提前熟悉好各个细节



样式



样式



- 预备突发情况

最好随便找一篇文章提前预演一下

交叉引用

➤	Assumf	[1] 参考文献	Figure 1: 图片	图表标题	正文
Table 1: 表格	1 标题 1	1.1 标题 2	1.1.1 标题 3	标题	
副标题	不明显强调	强调	明显强调	要点	
引用	明显引用	不明显参考	明显参考	书籍标题	
列表段落	题注	TOC 标题			
A+ 创建样式(S)					
A- 清除格式(C)					



Page 2 of 2

Contents	
1 Introduction	3
1.1 Problem Background	3
1.2 Restatement of the Problem	3
1.3 Our Work	4
2 Assumptions and Justifications	4
3 Notations and Data	5
4 Problem 1: Agricultural Ecosystem Based on Differential Equation	6
4.1 Problem analysis	6
4.2 Model preparation	6
4.3 Model 1: Dynamic Model of Tri-trophic Food Chain	7
4.4 Model 2: Dynamic Model of Food Web	8
4.5 Model 3: Food Web Model Considering Agricultural Cycles	13

论文写作 —— 摘要（画龙点睛）

三个细节：

- 一页以内
- 适当加粗

模型名称、直观数据、核心结论

- 标题、关键词

简单直接，题目的扩充

主体部分，根据
模型/问题来分段

敏感性分析与总结

Problem Chosen
E

2025
MCM/ICM
Summary Sheet

Team Control Number
2502355

Forests and Agriculture in Harmony: How to Maintain Ecosystem Stability During the Transition to Organic Farming Summary

Forests, often referred to as the "lungs of the Earth" and home to countless species, have been increasingly deforested in recent years. The cleared forests become converted forest areas, raising a critical challenge: how can we best utilize this land for sustainable purposes?

For Problem 1, we mainly employ differential equations and develop the equations progressively. We begin by constructing a simple food chain model, gradually advancing to a more complex food web model. This includes the introduction of pesticide and herbicide, followed by the incorporation of agriculture cycle and its seasonality into the model. The final model considers various factors such as **temperature**, **fertilization**, and the **periodic application of pesticides and herbicides**. By using numerical simulating the dynamics of the food web over the course of one year, we analyze the biomass changes of the species involved. As shown in Figure 9, the impact of agricultural cycles and seasonality on the system: Crop-like producers, grass-like producers, and primary consumers undergo **periodic oscillations** in response to the periodic application of pesticides and herbicides.

For Problem 2, a network is established using graph theory, with populations as nodes. The carbon flow, calculated based on the biomass from the equilibrium simulations in Problem 1, serves as the arc weight. Different network scenarios are constructed by sequentially incorporating factors such as the reemergence of two species, the application of pesticides and herbicides, and the inclusion of bats and bees. The **stability**, **biodiversity**, and **centrality** indicators of the network are then evaluated and assessed, providing a comprehensive understanding of the system's dynamics. Bats (**29.29**) and Bees (**43.37**) both result in greater stability, with the removal of pesticides and herbicides and doing nothing yielding a stability of **56.99**.

For Problem 3, we propose three specific measures to achieve organic farming: improving soil nutrients, pest management, and regular manual weeding. These measures are tested in pairs, all three combined, and a blank control solution. We evaluate their effectiveness by **numerically simulating** the differential equations from Problem 1 to determine the stable biomass of each species under each solution. Using data from the differential equations, calculated metrics, and expert evaluations, we analyze economic, ecological, sustainability, and social benefits. An **EWM-AHP** model is developed to assess these schemes for organic farming. Based on the results, we recommend the optimal solution for farmers: improving soil conditions combined with regular manual weeding.

Finally, a sensitivity analysis of the herbicide power (alpha value) in our Problem 1 food web model shows that the **selected alpha value** is reasonable and appropriate for the simulation. Based on the ranking results of the methods combination for organic farming, we drafted a one-page letter offering clear and practical suggestions for farmers looking to transition to organic farming. These suggestions carefully balance economic trade-offs with sustainability considerations, ensuring that the advice is both actionable and aligned with long-term ecological and economic goals.

Keywords: Differential equation, Food web model, network, EWM-AHP

03

参赛经历

Competition Experience

赛前

编程手 备赛

1-模型及代码

模型原理，代码全部准备好
队友也一起下载python、vscode

2-过往O奖论文研读

比如2024年的A题论文的一些绘图

3-运筹学，伪代码 etc.

如果有时间，可以再学一些别的基础的

Algorithm 1 Lamprey Population Iteration Simulation

Input:

The maximum iterations, G
Initial amount of larval fish l_1
Initial amount of adult male fish l_2^{male}
Initial amount of adult female fish l_2^{female}

Output:

Larval fish amount in G year(s) l_1
Adult fish amount in G year(s) l_2
Total amount in G year(s) l
Gender ratio in G year(s) gk

```
1: for  $i \leftarrow 1 : G$  do
2:   Calculate the decrease amount of  $l_2^{male}$ 
3:   Calculate the decrease amount of  $l_2^{female}$ 
4:    $b \leftarrow \frac{b_0}{(1 + \ln^2 \frac{l_2^{female}(l_1 + l_2)}{l_2^{male} K})}$ 
5:    $mate \leftarrow round((l_2^{male} + l_2^{female})b + rand())$ 
6:    $l_2^{male} = l_2^{male} - mate$ 
7:    $l_2^{female} = l_2^{female} - mate$ 
8:   for  $j \leftarrow 1 : 100$  do
9:     Calculate the decrease amount of  $l_1^j$ 
10:    Calculate the amount of fish that grows into adulthood  $change$ 
11:     $p \leftarrow 1 - pow2(\alpha + \beta j)$ 
12:     $male \leftarrow round(change * p + rand())$ 
13:     $l_2^{male} = l_2^{male} + male$ 
14:     $l_2^{female} = l_2^{female} + change - male$ 
15:  end for
16:  for  $j \leftarrow 99 : -1 : 1$  do
17:     $l_1^{j+1} = l_1^j$ 
18:  end for
19:   $l_1(i) \leftarrow sum(l_1)$ 
20:   $l_2(i) \leftarrow l_2^{male} + l_2^{female}$ 
21:   $l(i) \leftarrow l_1(i) + l_2(i)$ 
22:   $gk(i) \leftarrow \frac{l_2^{male}}{l_2(i)}$ 
23: end for
```

赛中
100h

1/24

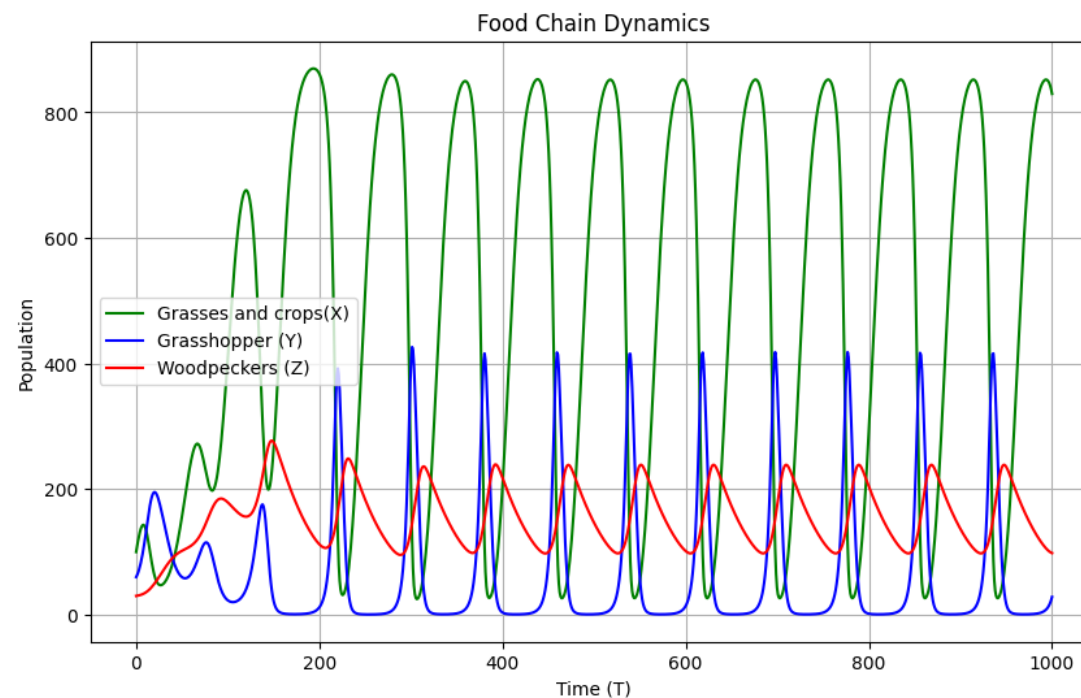
- 上午 选题
- 下午晚上 模型选择，背景资料，看论文
- 初步确定了食物链，相关模型，初步代码（对应论文的 1 2 3 部分）
- 时间：上午9:00-凌晨11:35

test.ipynb

2025/1/24 18:29

Jupyter 源文件

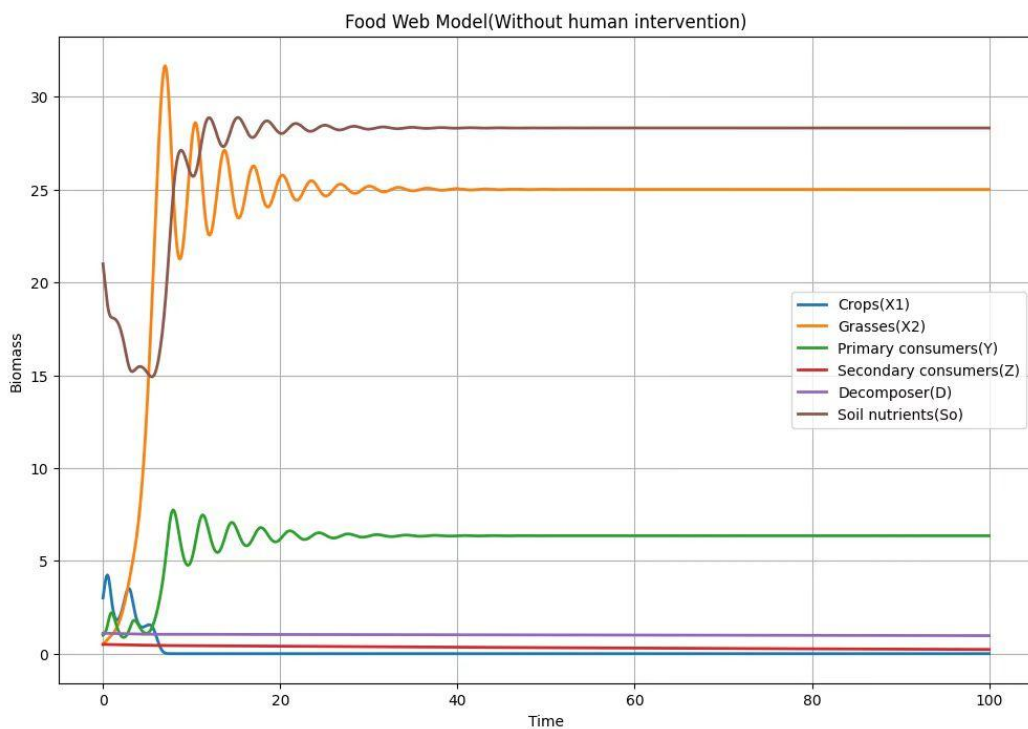
365 KB



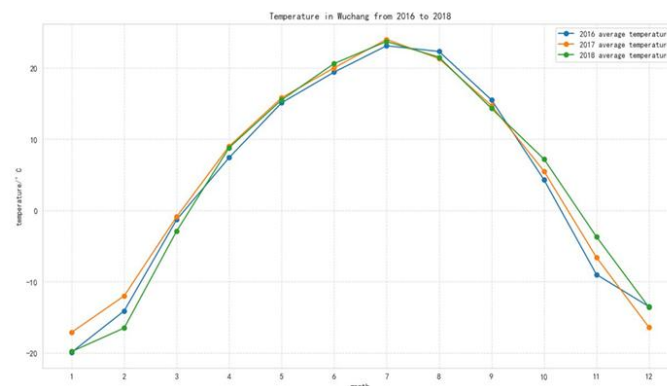
赛中
76h

1/25

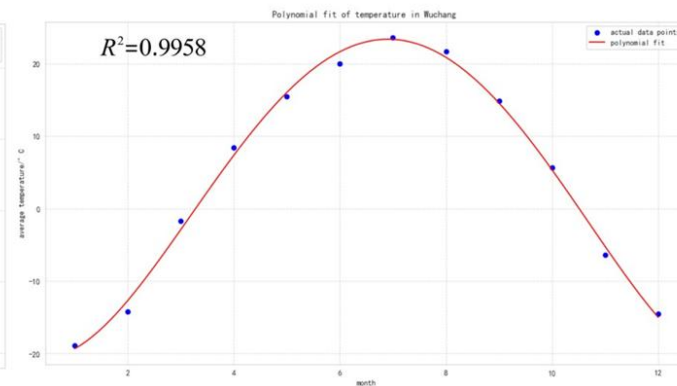
- 早中晚 数值模拟，绘图，继续阅读论文
- 晚上 水稻生长，气温数据
- 4.3 Model 1: Dynamic Model of Tri-trophic Food Chain
- 4.4.1 Food web model (without human intervention)
- 时间：上午9:30-凌晨12:40



$$T = -4.2071g(t) + 5.1125g(t)^2 - 0.6947g(t)^3 + 0.0250g(t)^4 - 19.57054 \quad (13)$$



(a)



(b)

赛中
52h

1/26

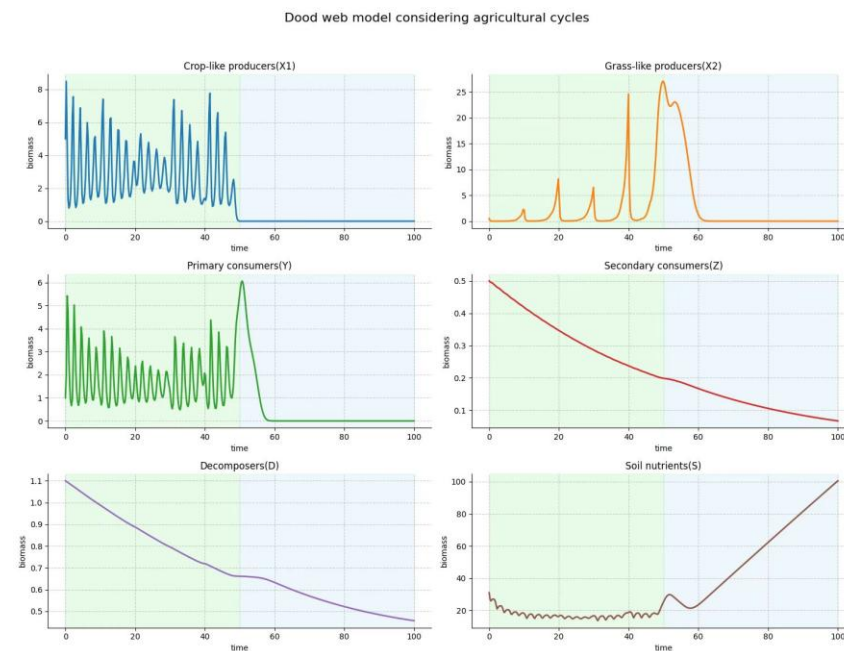
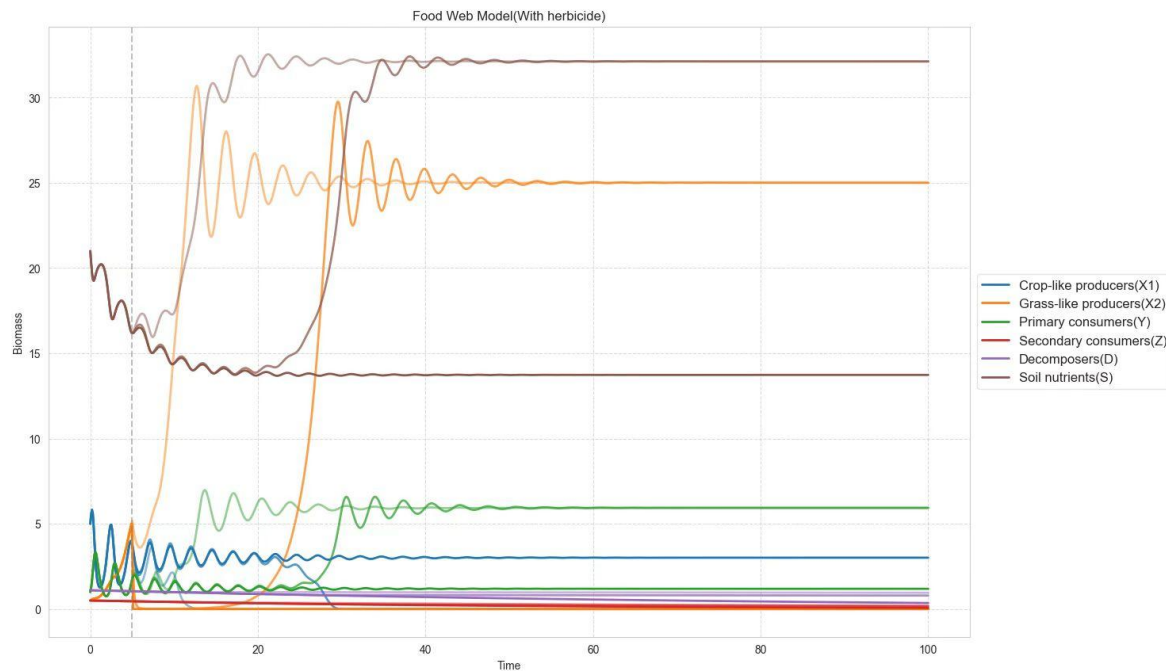
- 下午 主要是除草和杀虫的不同比例绘图 (4.4.2)
- 晚上 不同季节变化 物种变化 (4.5)
- 4.4.2 Food web model (with human intervention)
- 4.5 Model 3: Food Web Model Considering Agricultural Cycles
- 时间: 上午9:40-凌晨1:22

除草和杀虫.docx

2025/1/26 16:28

DOCX 文档

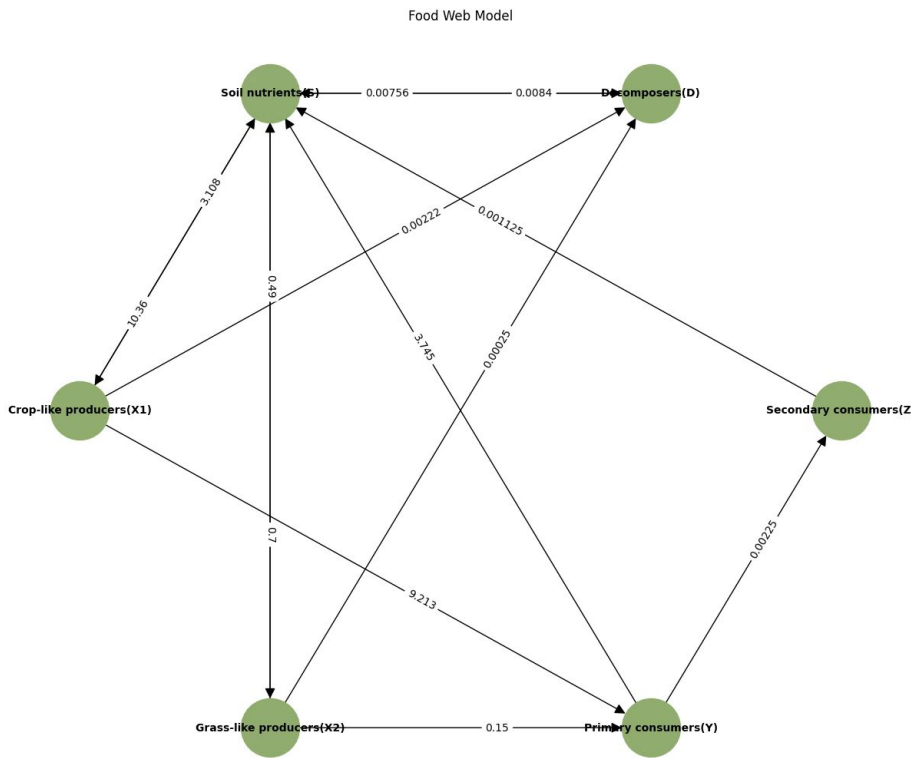
1,705 KB



赛中
28h
生产力大爆发!

1/27

- 上午下午 开始图论，绘制食物网，计算各项指标（对应5）
- 晚上 评估各种方案（对应6），继续计算节点指标（5）
- 5 Problem 2: Ecosystem Assessment
- 6 Problem 3: Towards Green Agriculture
- 时间：上午10:00-1/28



25/1/27 11:44

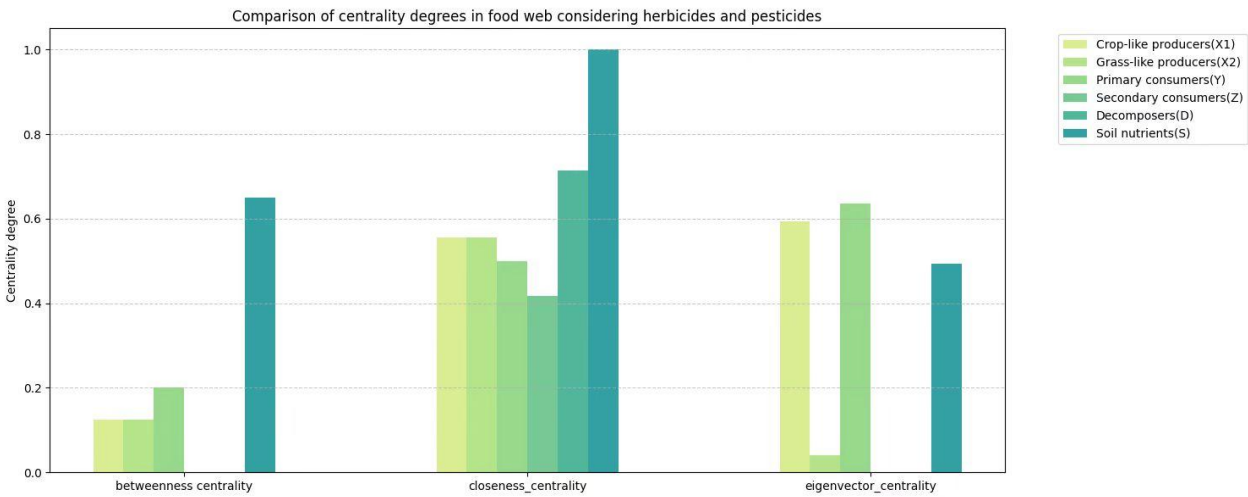
Jupyter 源文件

537 KB

25/1/27 13:10

Jupyter 源文件

936 KB



25/1/27 23:25

XLSX 工作表

16 KB

25/1/27 23:30

Jupyter 源文件

12 KB

25/1/27 23:49

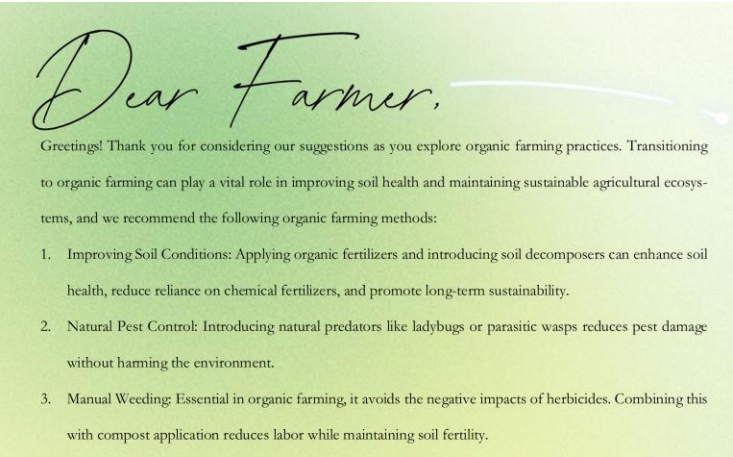
Jupyter 源文件

1,559 KB

赛中
9h
生产力大爆发!

1/28

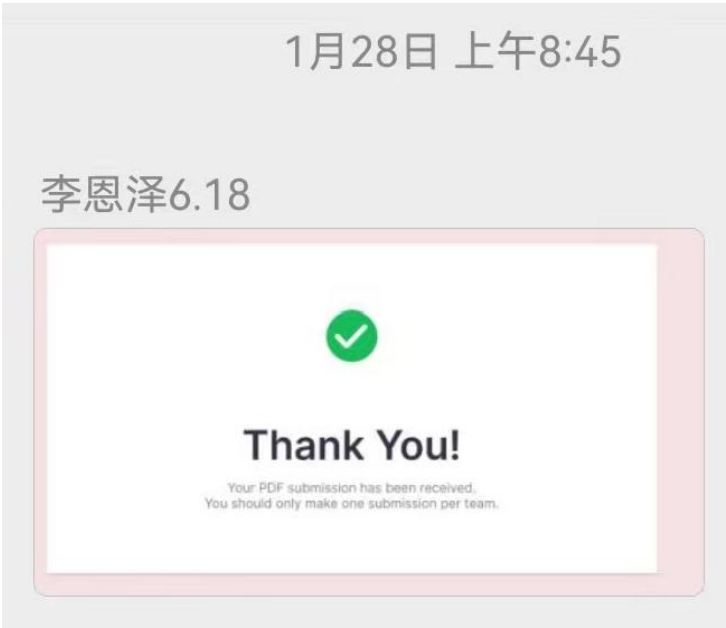
- 凌晨1:31 完成敏感性分析，及对应部分论文
- 凌晨2点 翻译论文中
- 凌晨3点 写最后的给farmer的信
- 凌晨4点-5点 写摘要
- 凌晨6点我去休息了，但是队友们奋战到了9点多（论文超出面数了）
- 8:45 提交



Summary

Forests, often referred to as the "lungs of the Earth" and home to countless species, have been increasingly deforested in recent years. The cleared forests become converted forest areas, raising a critical challenge: how can we best utilize this land for sustainable purposes?

For Problem 1, we mainly employ differential equations and develop the equations progressively. We begin by constructing a simple food chain model, gradually advancing to a more complex food web model. This includes the introduction of pesticide and herbicide, followed by the incorporation of agriculture cycle and its seasonality into the model. The final model considers various factors such as **temperature**, **fertilization**, and the **periodic application of pesticides and herbicides**. By using numerical simulating the dynamics of the food web over the course of one year, we analyze the biomass changes of the species involved. As shown in Figure 9, the impact of agricultural cycles and seasonality on the system: Crop-like producers, grass-like producers, and primary consumers undergo **periodic oscillations** in response to the periodic application of pesticides and herbicides.



赛后

总结优势

1-完整性

不要虎头蛇尾！
美赛是面向应用实践的：6 Problem 3: Towards Green Agriculture

2-新颖性，变化性

跨学科交叉
4 Problem 1: Agricultural Ecosystem Based on Differential Equation
5 Problem 2: Ecosystem Assessment

3-teamwork!

队友靠谱、脾气好，赛前一个月至少得每周开开会讨论一下，把整个学完

4-美工很重要，但不是everything。内容为王

Problem Chosen	2025	Team Control Number
E	MCM/ICM	2502355
	Summary Sheet	

Forests and Agriculture in Harmony: How to Maintain Ecosystem Stability During the Transition to Organic Farming

04 | 总结寄语

Best Wishes

经验谈

1-赛前充分准备

2-赛中放平心态，相信队友
情绪稳定

3-如果没有最初按照的时间安排进行，无需紧张，都能做完

4-引人入胜的标题、摘要很重要！这是评审第一关。

Problem Chosen
E

2025
MCM/ICM
Summary Sheet

Team Control Number
2502355

Forests and Agriculture in Harmony: How to Maintain Ecosystem Stability During the Transition to Organic Farming

经验谈

1-建模手所学专业

不一定需要数学专业，数理基础比较好的专业，例如物理、电信、电气、计算机类（人工智能）等，都适合充当建模角色

2-从高质量论文中建模

广泛搜索阅读文献，扩大关键词选择范围，先速览再精读

不要局限于（照搬）数模课程中学习的模型

3-没有绝对分工，只有相对擅长

比赛过程中经常需要身兼数职，可以分配主要职责+次要任务。勇于承担，不要斤斤计较。

4-有效沟通，高效会议

无论线上线下参赛，有问题、有想法、有意见及时提出，队友间平和沟通，少数服从多数。

5-团队组建

队友间最好相互认识，效果往往比 qq群/小红书 组队效果好
有共事经历（小组作业/高中同学/社会实践）的同学组队，合作氛围更好
队友尽量来自不同学院，有助于能力互补和灵感碰撞

经验小结

1-一定要多和建模手多沟通建模思路

在动笔之前更重要的是要先与建模手理清思路和逻辑
不耻下问，只要没搞清楚就直接问

2-注重细节，精益求精

画图、格式中的很多细节

3-前期工作量较小的时候多看往年类似论文

作为论文手，前期的工作量会相对来说小一点点
可以多看往年类似论文并进一步加强文章逻辑的构建

4-心态调节，合理定位

作为化学专业，对数模相对陌生
但不需要妄自菲薄，重要的还是逻辑（而且有的时候也可以提供一些新的视角）

5-相信自己

比赛过程中会有很多的外界信息，但一定要相信自己，决定好了就一路做下去！

祝愿大家取得理想结果！